

ML Hybrid Models Comparison Report

Generated: 2025-11-23T22:42:26.027558

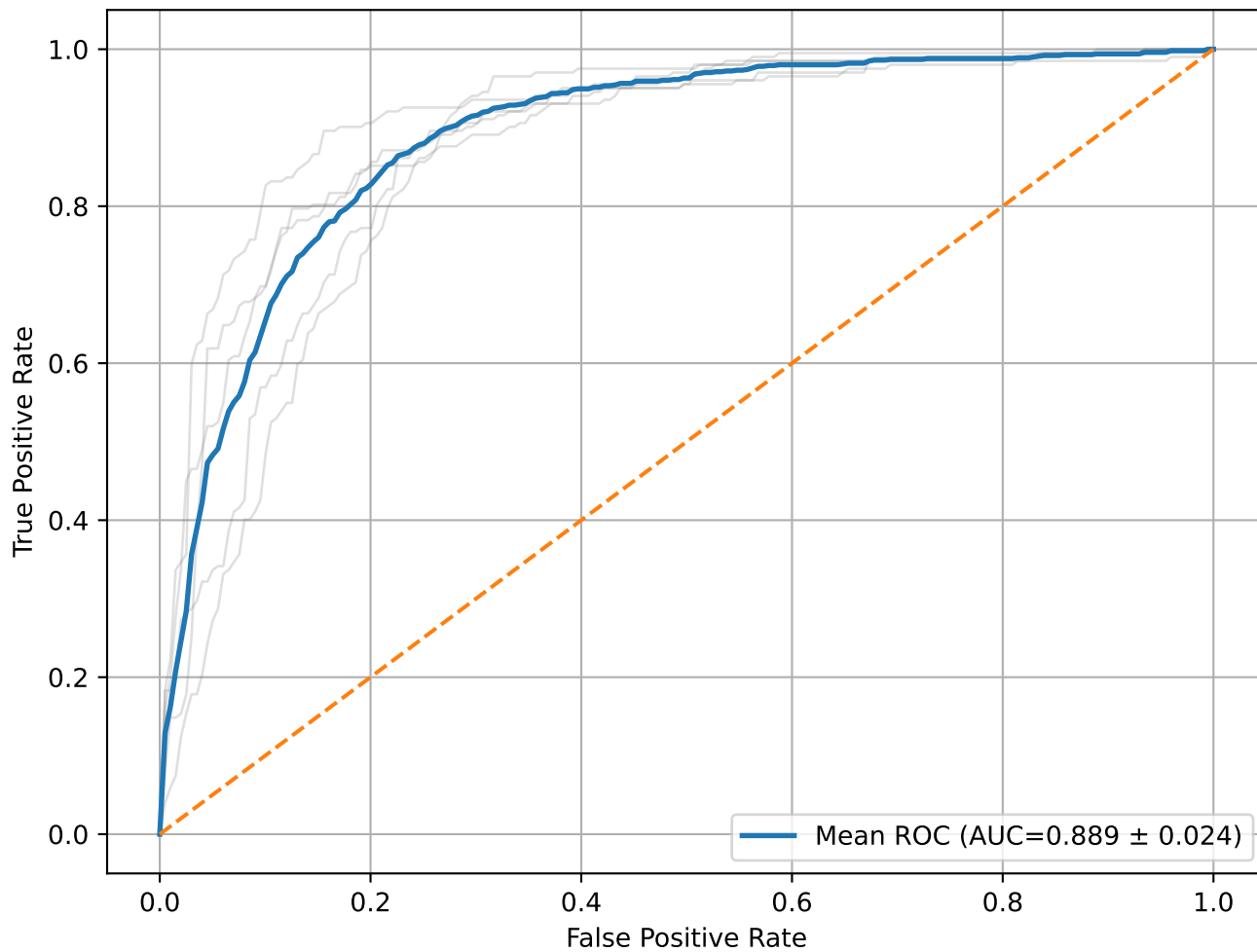
Dataset samples: 2009, features: 447

Models: logistic, rf, svm, xgb, lgb, dnn, stacking, voting, bagging, pca_svm, ga_xgb, pso_svm, bayes_opt_svms

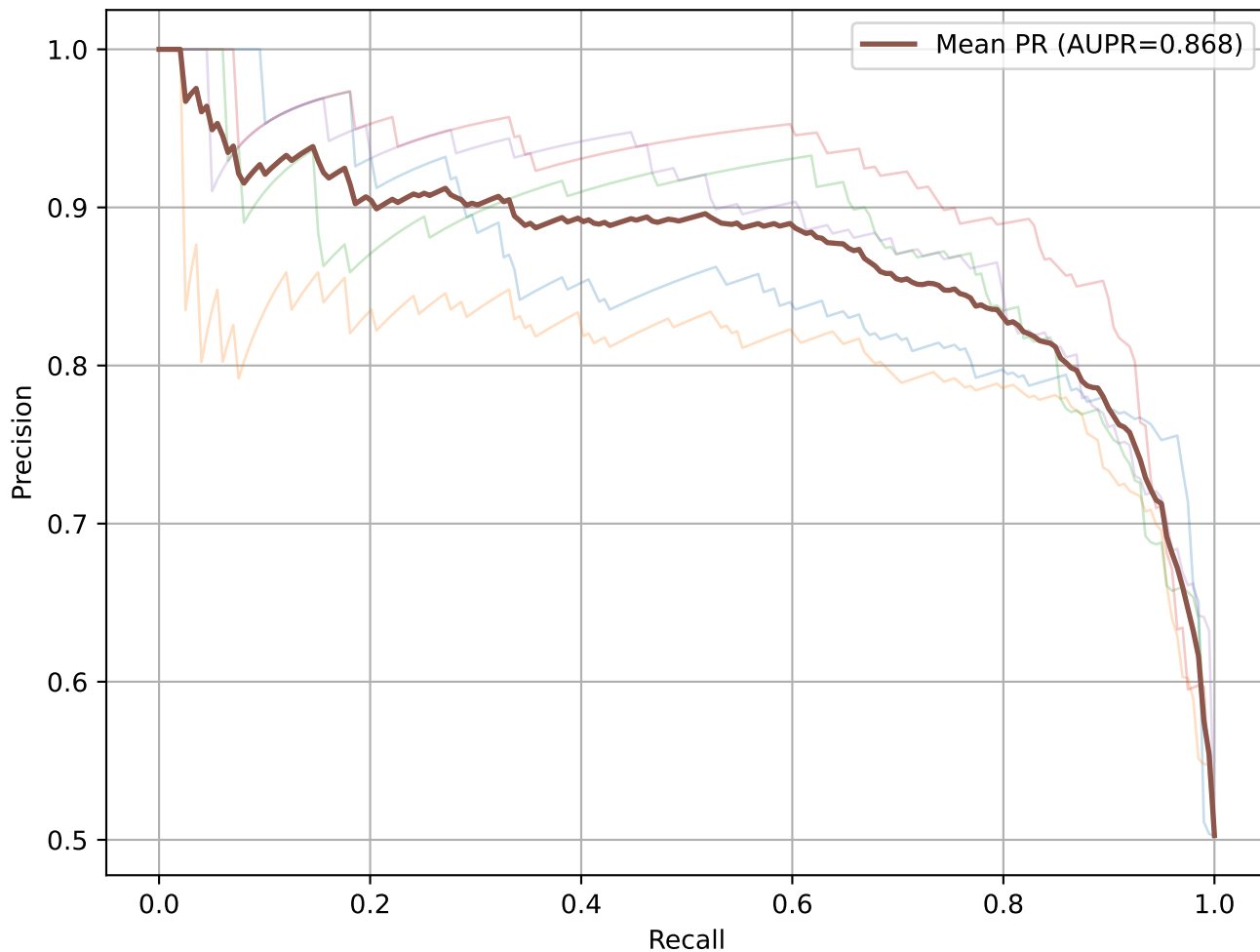
StratifiedKFold: 5 splits

Random seed: 42

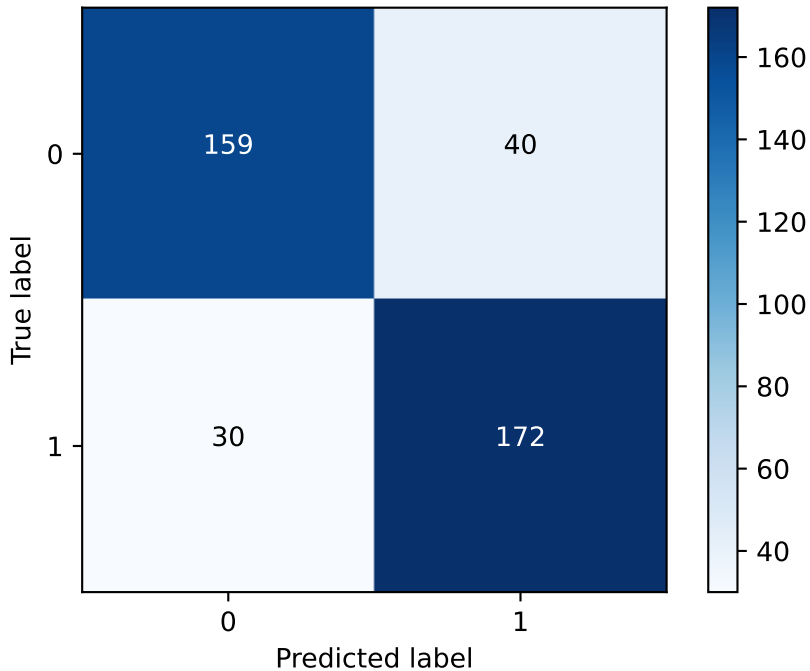
ROC - LOGISTIC



Precision-Recall - LOGISTIC



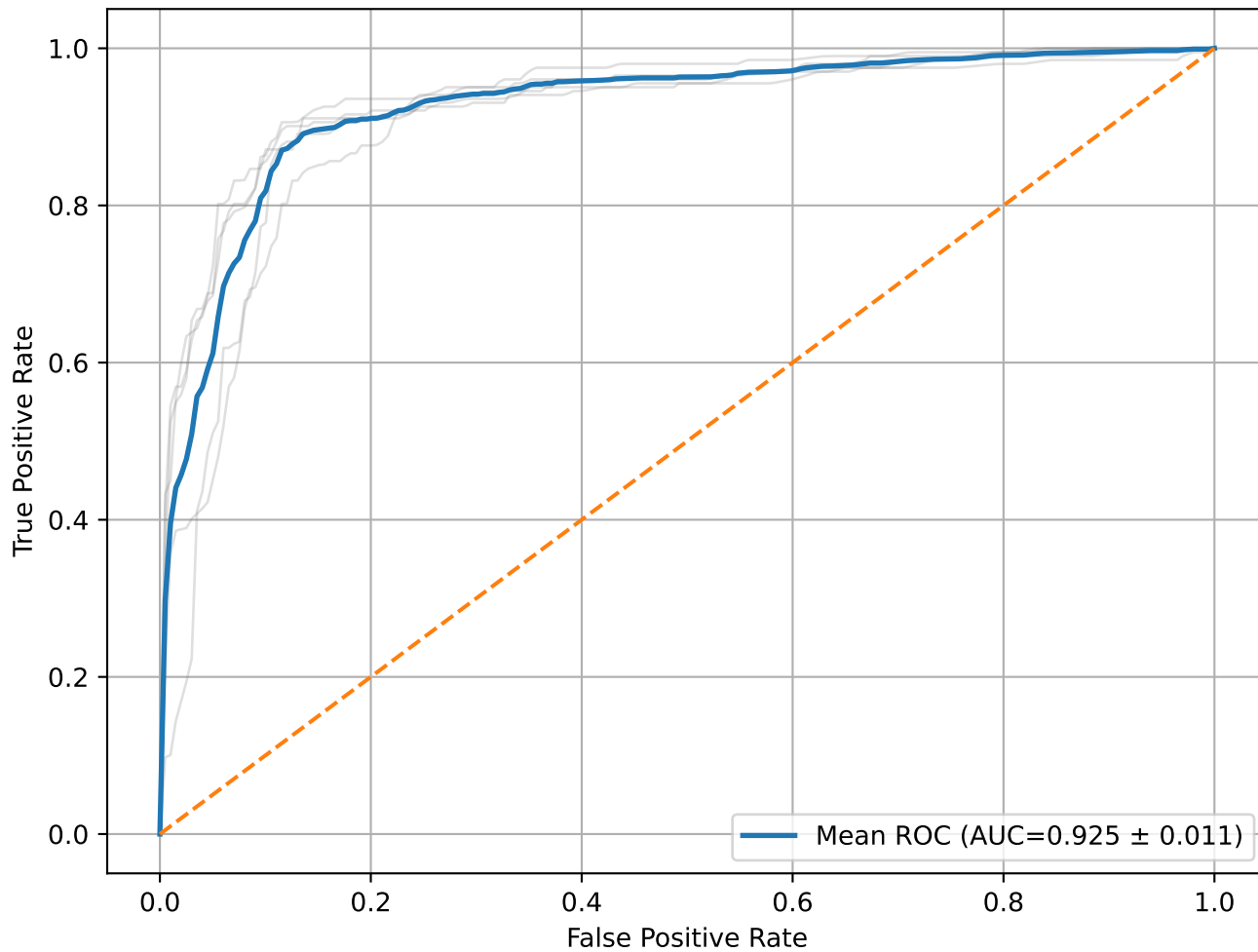
Confusion Matrix - LOGISTIC (last fold)



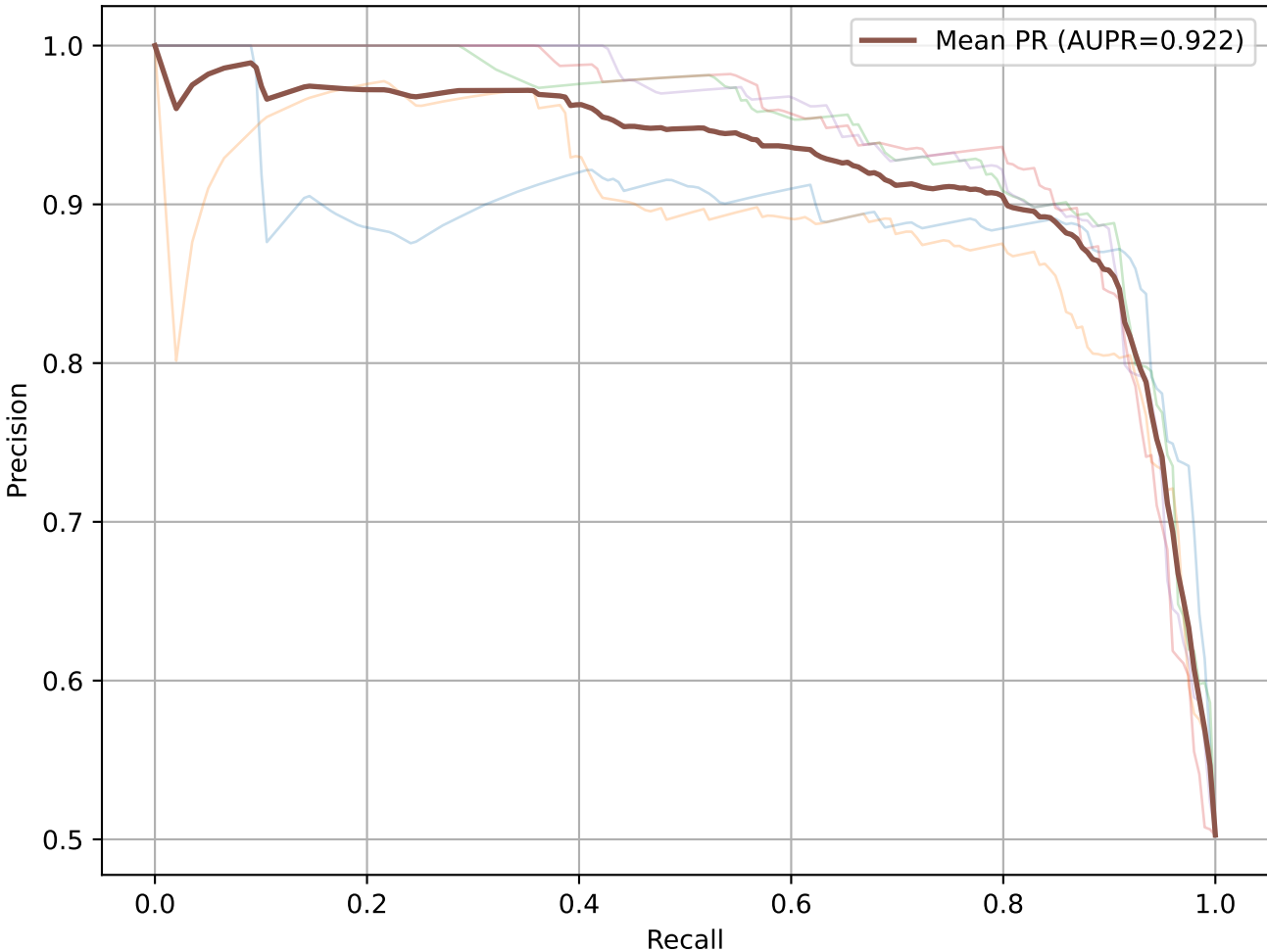
ClassificationReport - LOGISTIC (last fold)

	precision	recall	f1-score	support
0	0.84	0.80	0.82	199
1	0.81	0.85	0.83	202
accuracy			0.83	401
macro avg	0.83	0.83	0.83	401
weighted avg	0.83	0.83	0.83	401

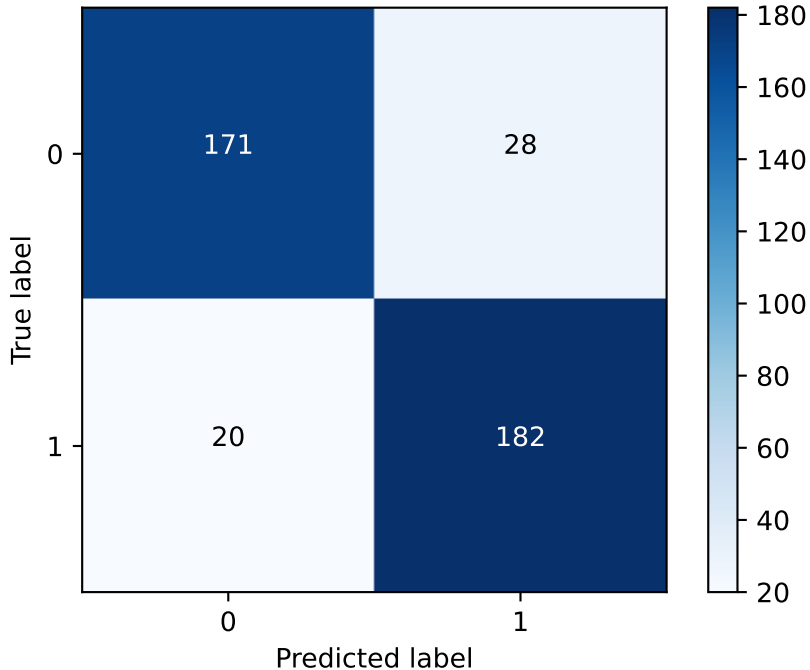
ROC - RF



Precision-Recall - RF



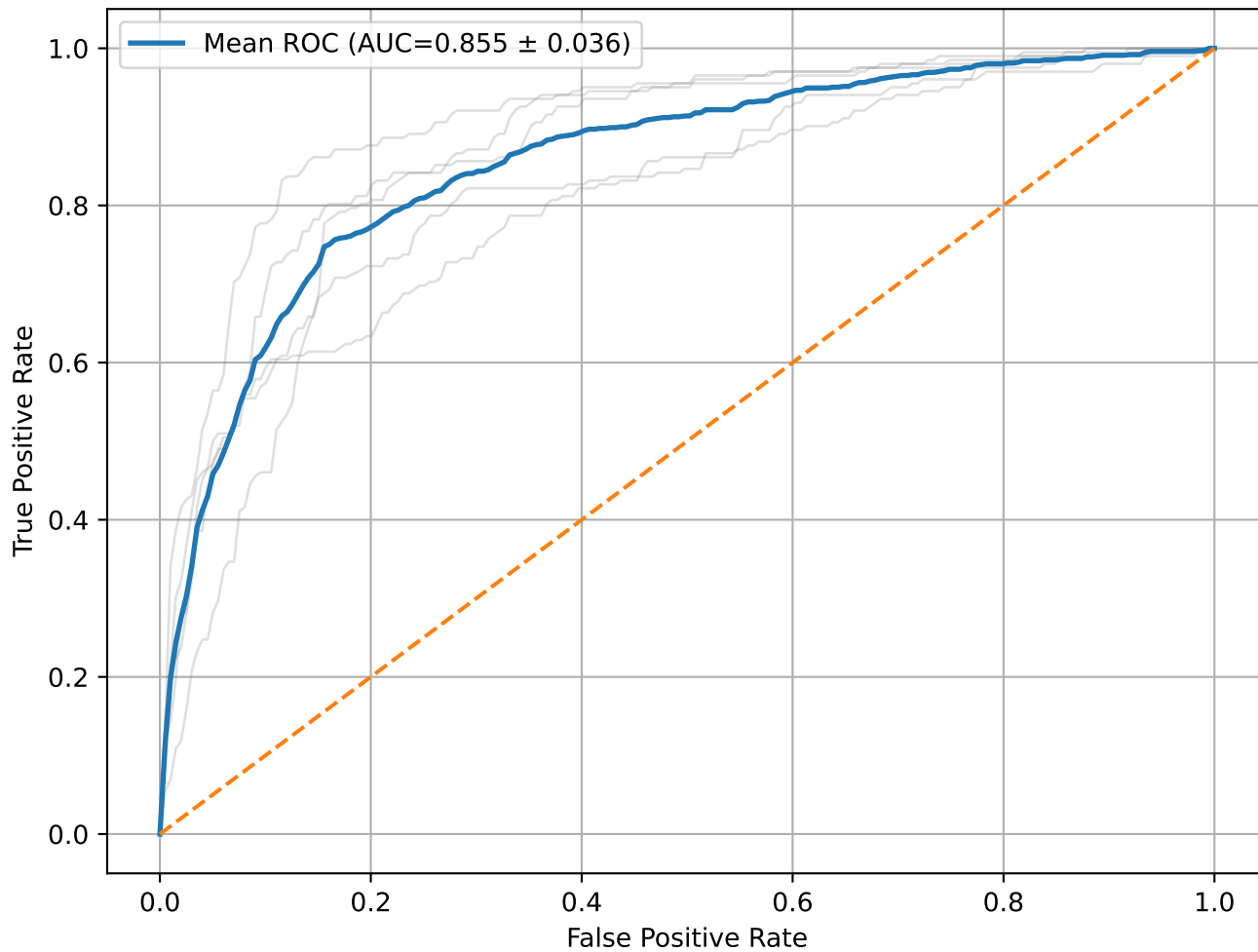
Confusion Matrix - RF (last fold)



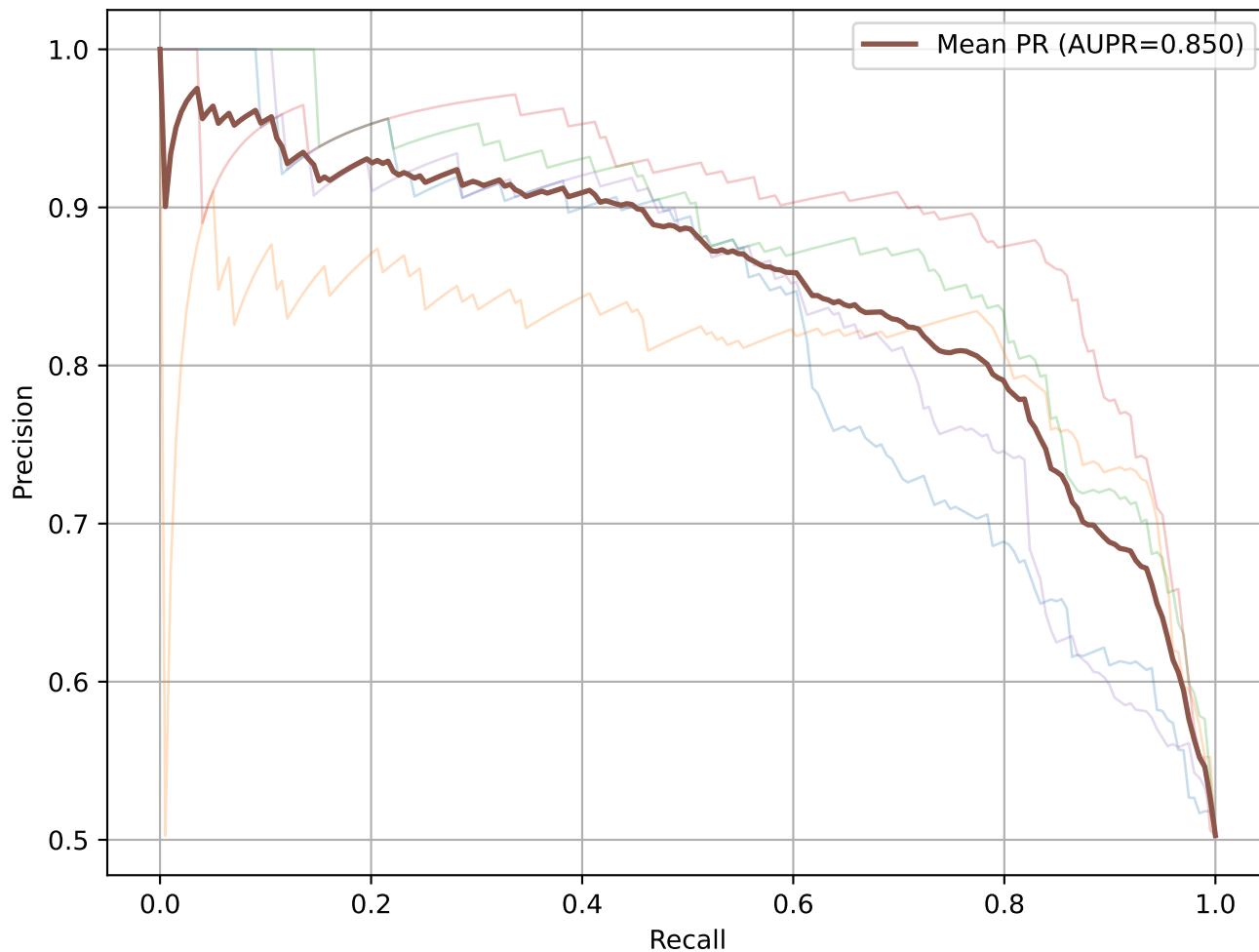
ClassificationReport - RF (last fold)

	precision	recall	f1-score	support
0	0.90	0.86	0.88	199
1	0.87	0.90	0.88	202
accuracy			0.88	401
macro avg	0.88	0.88	0.88	401
weighted avg	0.88	0.88	0.88	401

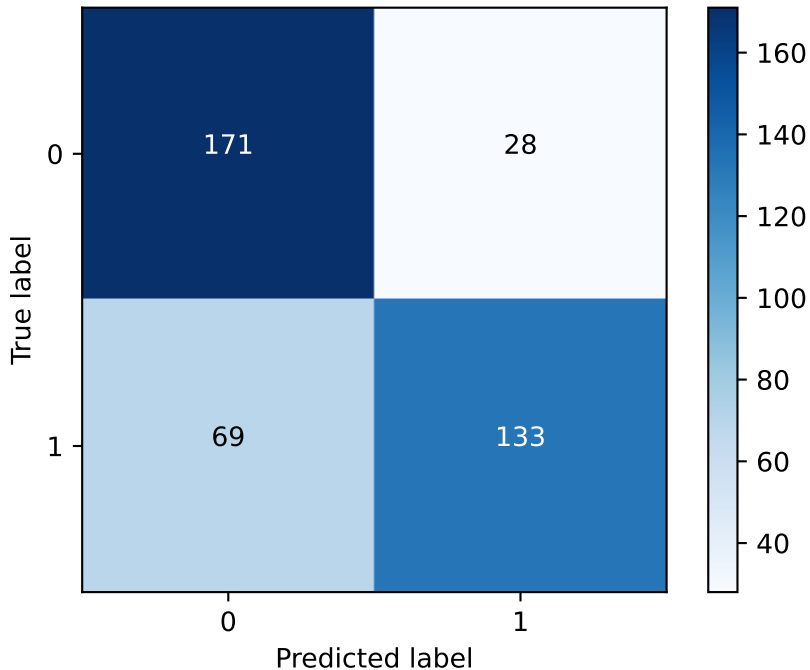
ROC - SVM



Precision-Recall - SVM



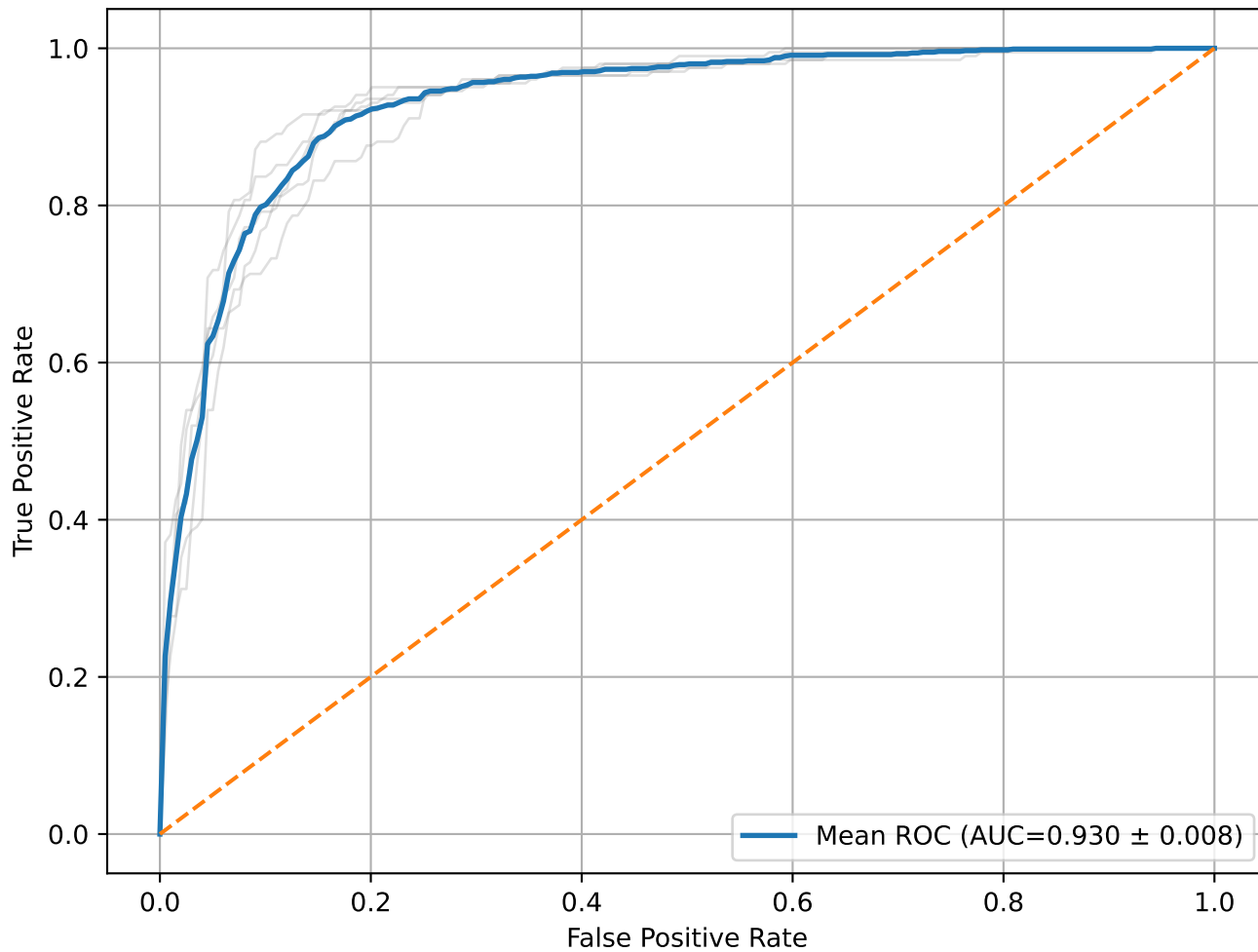
Confusion Matrix - SVM (last fold)



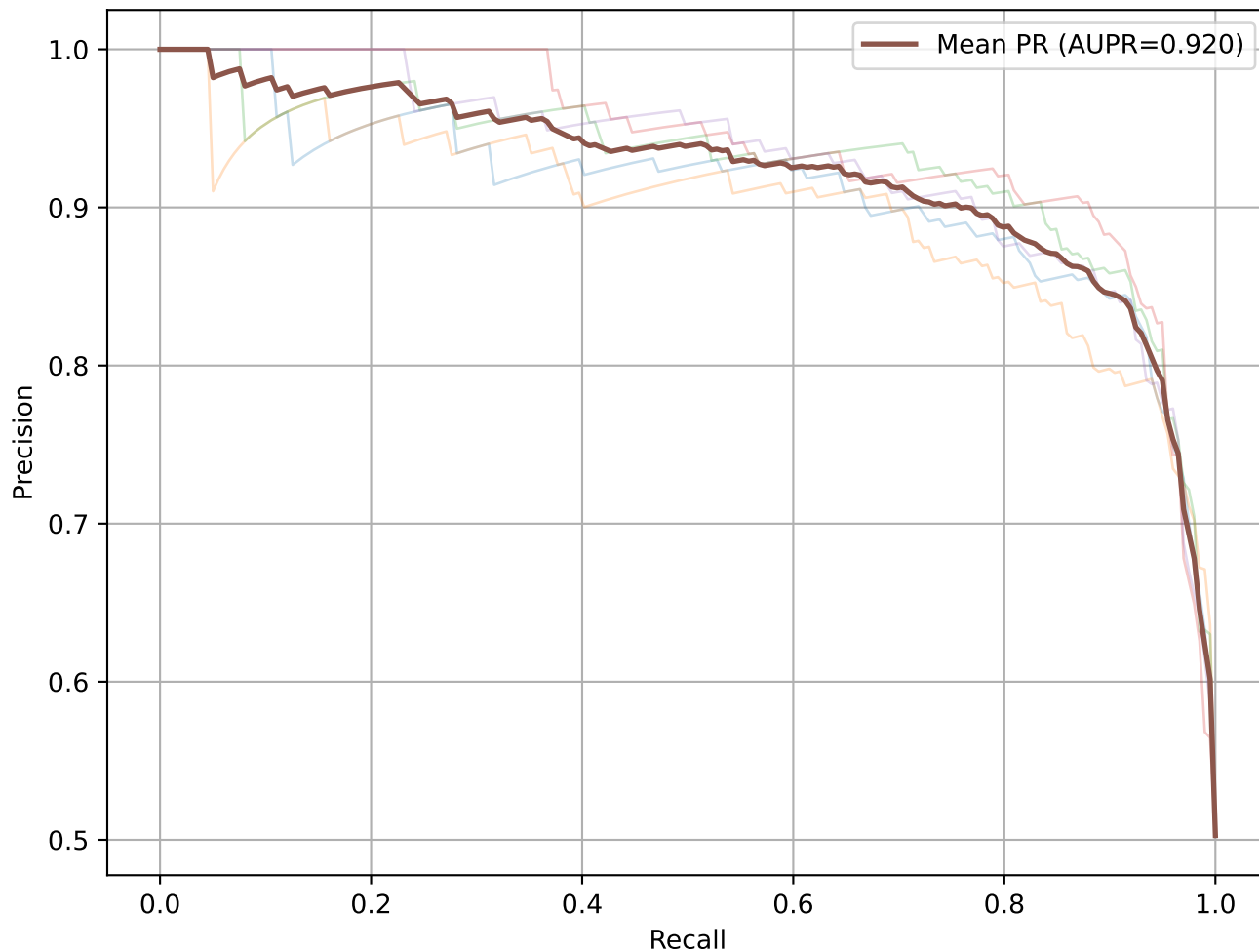
ClassificationReport - SVM (last fold)

	precision	recall	f1-score	support
0	0.71	0.86	0.78	199
1	0.83	0.66	0.73	202
accuracy			0.76	401
macro avg	0.77	0.76	0.76	401
weighted avg	0.77	0.76	0.76	401

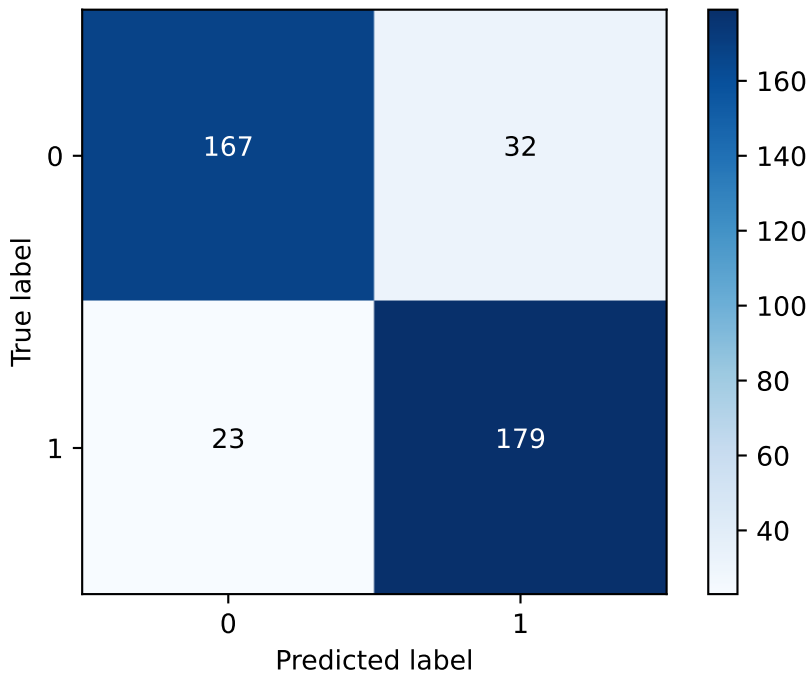
ROC - XGB



Precision-Recall - XGB



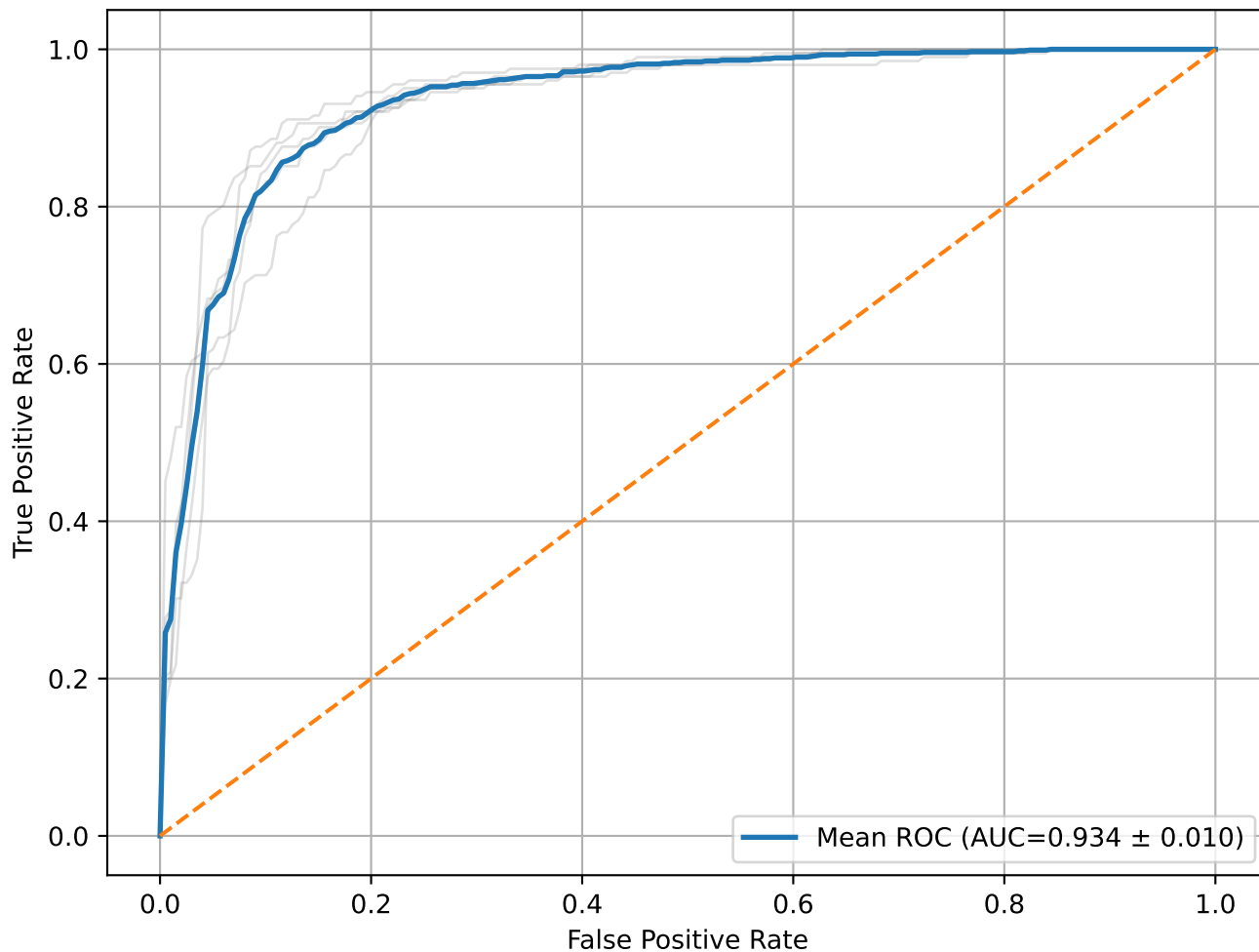
Confusion Matrix - XGB (last fold)



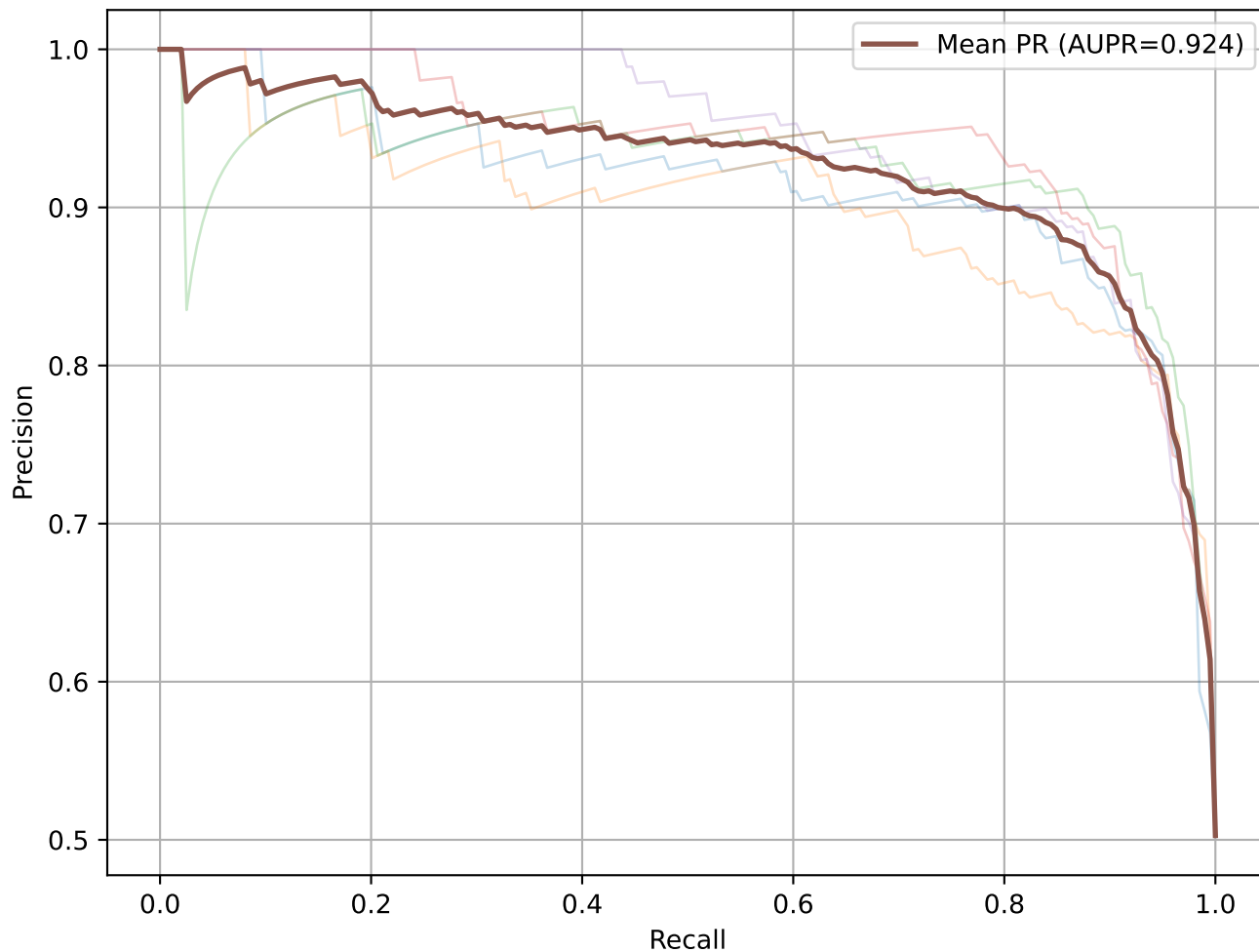
ClassificationReport - XGB (last fold)

	precision	recall	f1-score	support
0	0.88	0.84	0.86	199
1	0.85	0.89	0.87	202
accuracy			0.86	401
macro avg	0.86	0.86	0.86	401
weighted avg	0.86	0.86	0.86	401

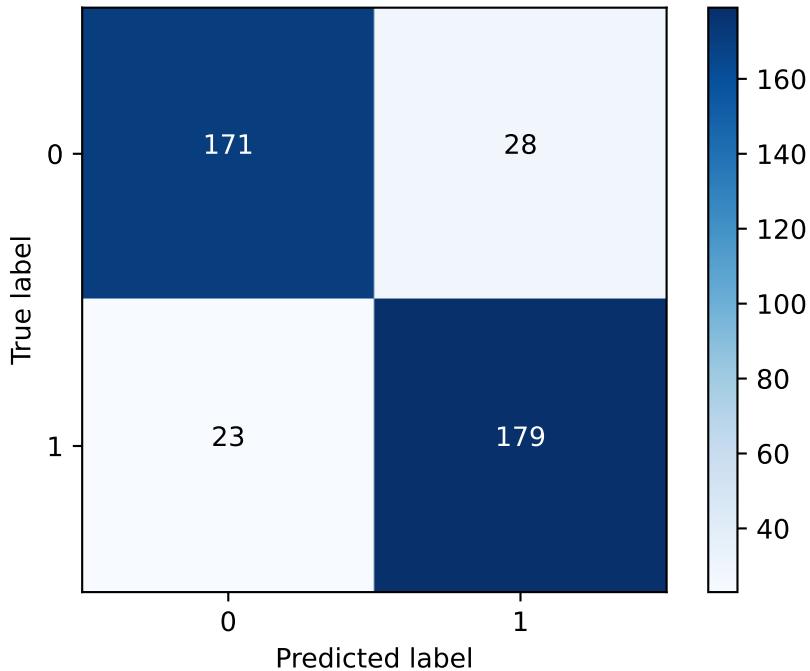
ROC - LGB



Precision-Recall - LGB



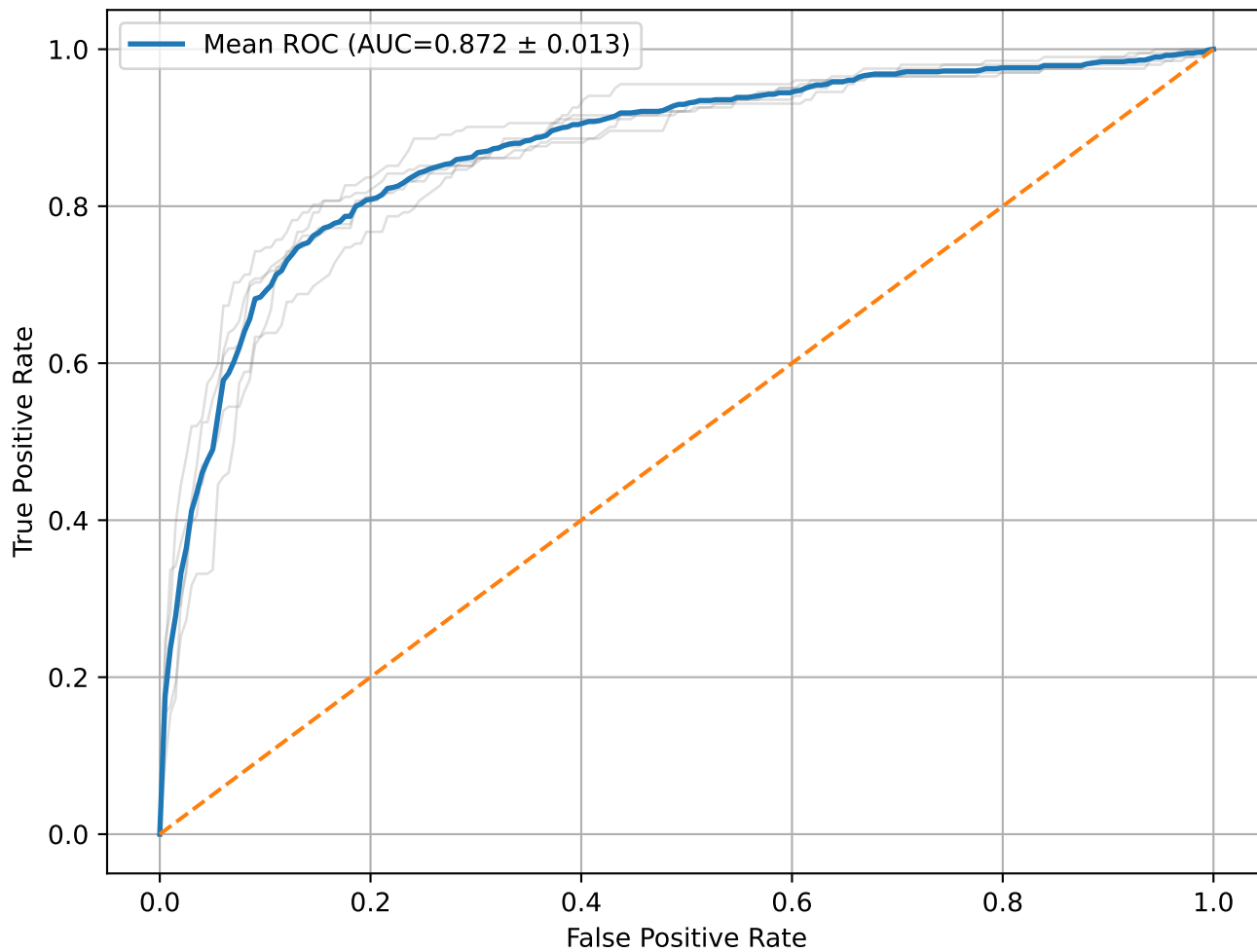
Confusion Matrix - LGB (last fold)



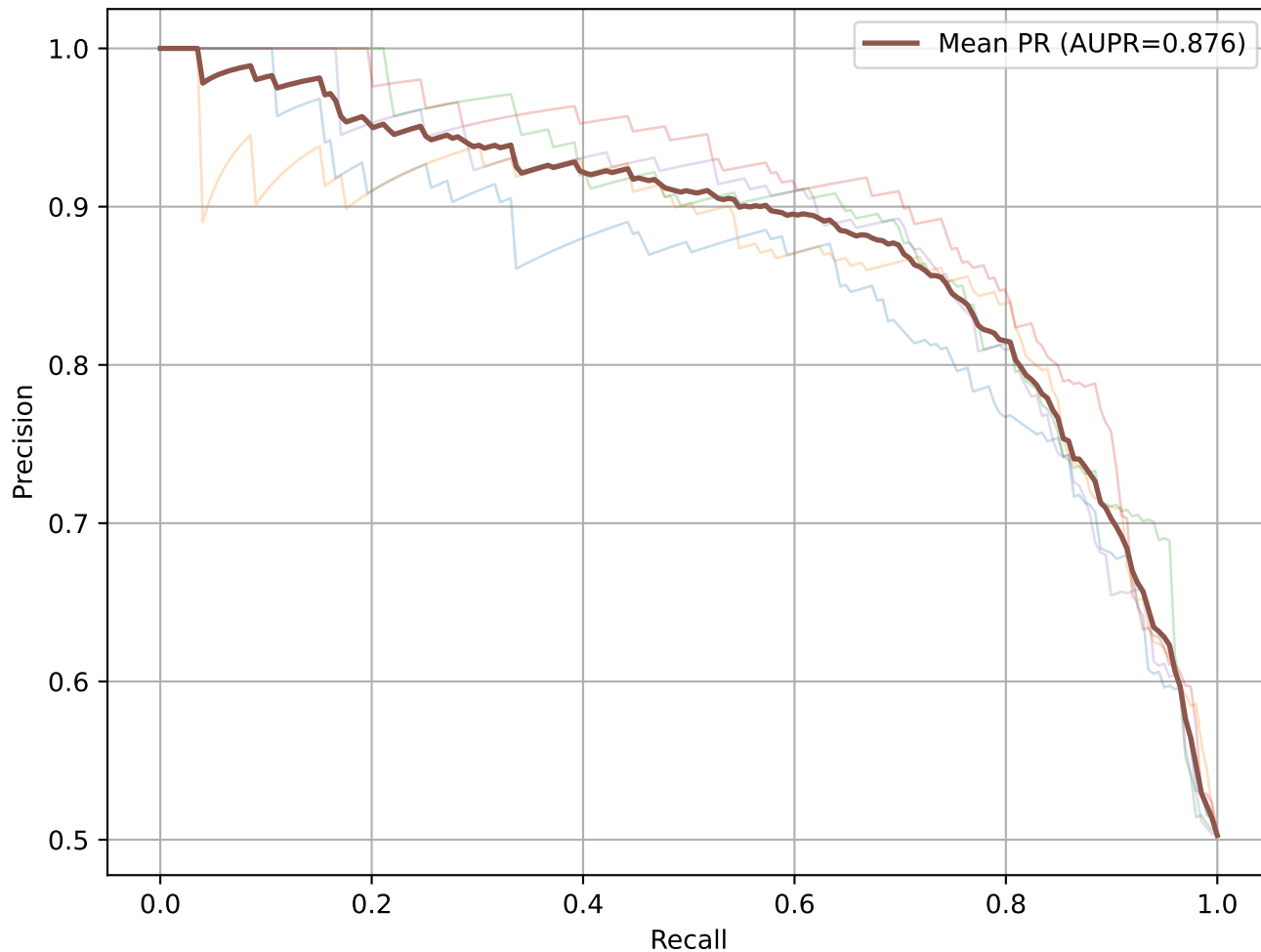
ClassificationReport - LGB (last fold)

	precision	recall	f1-score	support
0	0.88	0.86	0.87	199
1	0.86	0.89	0.88	202
accuracy			0.87	401
macro avg	0.87	0.87	0.87	401
weighted avg	0.87	0.87	0.87	401

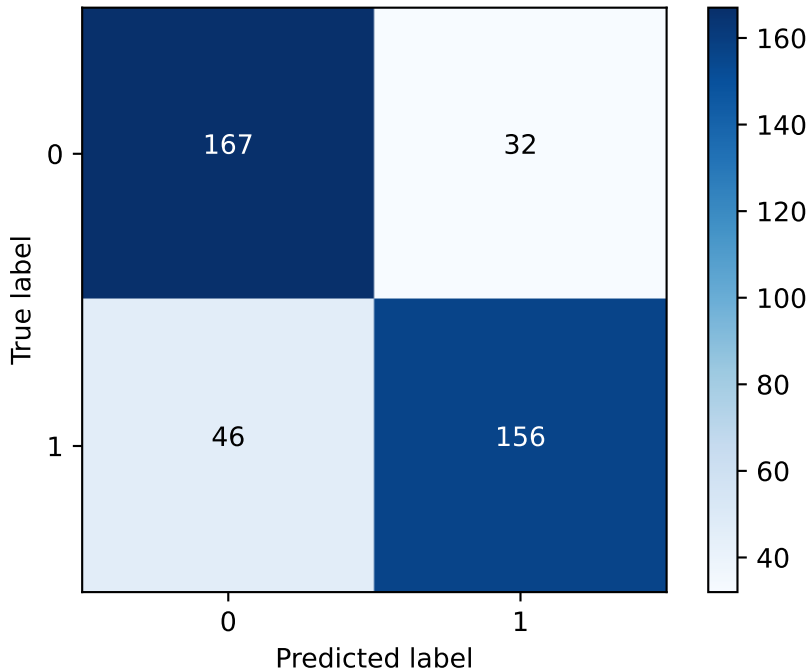
ROC - DNN



Precision-Recall - DNN



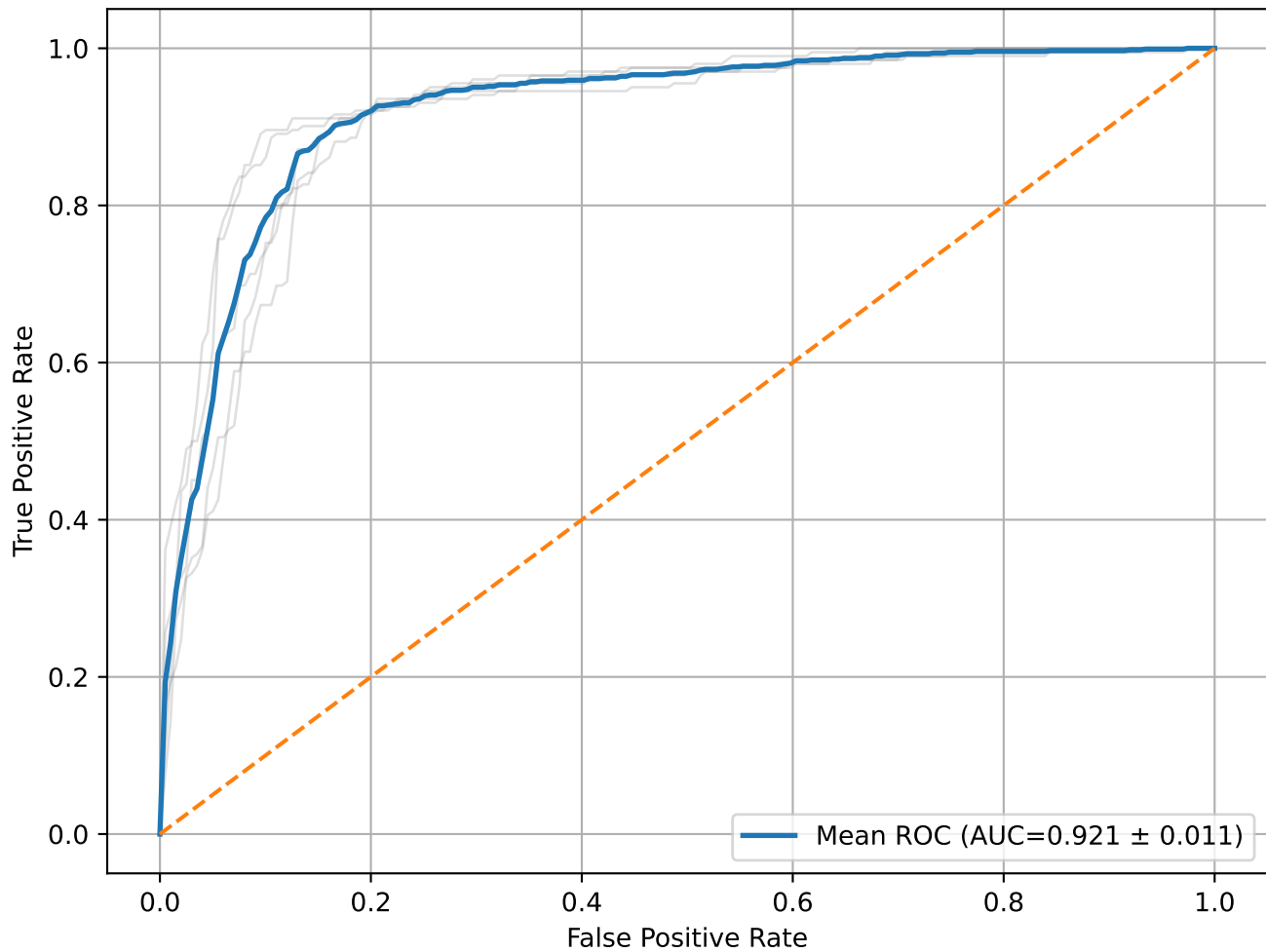
Confusion Matrix - DNN (last fold)



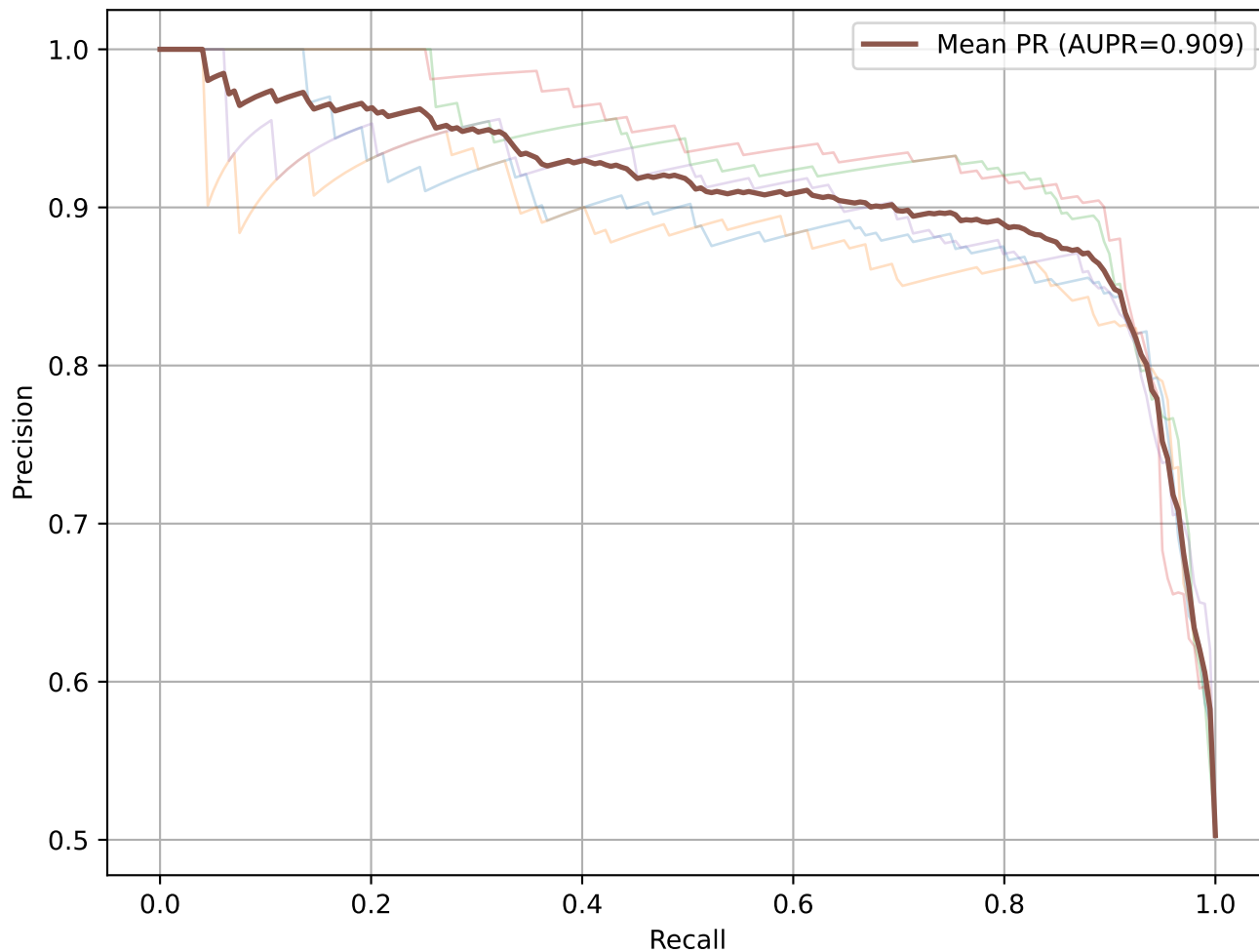
ClassificationReport - DNN (last fold)

	precision	recall	f1-score	support
0	0.78	0.84	0.81	199
1	0.83	0.77	0.80	202
accuracy			0.81	401
macro avg	0.81	0.81	0.81	401
weighted avg	0.81	0.81	0.81	401

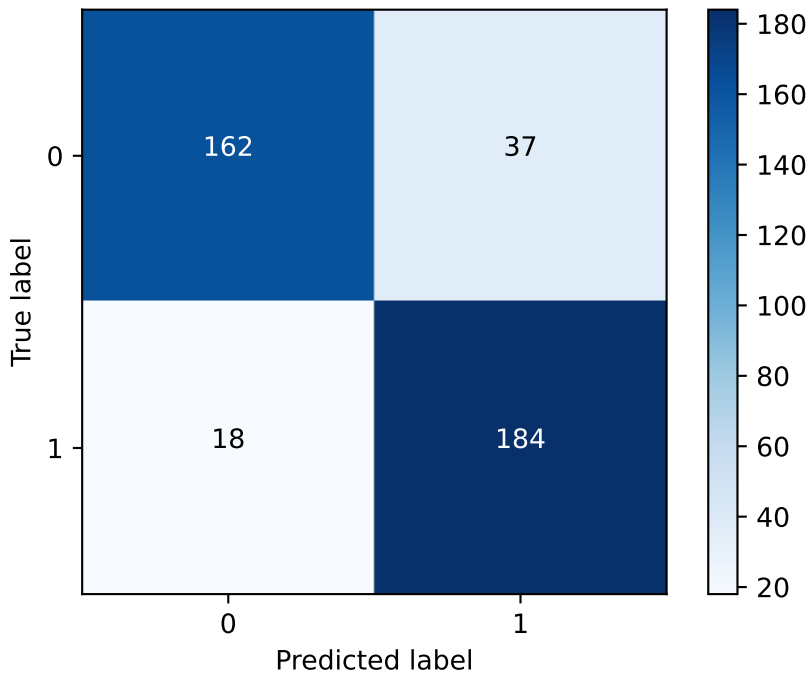
ROC - STACKING



Precision-Recall - STACKING



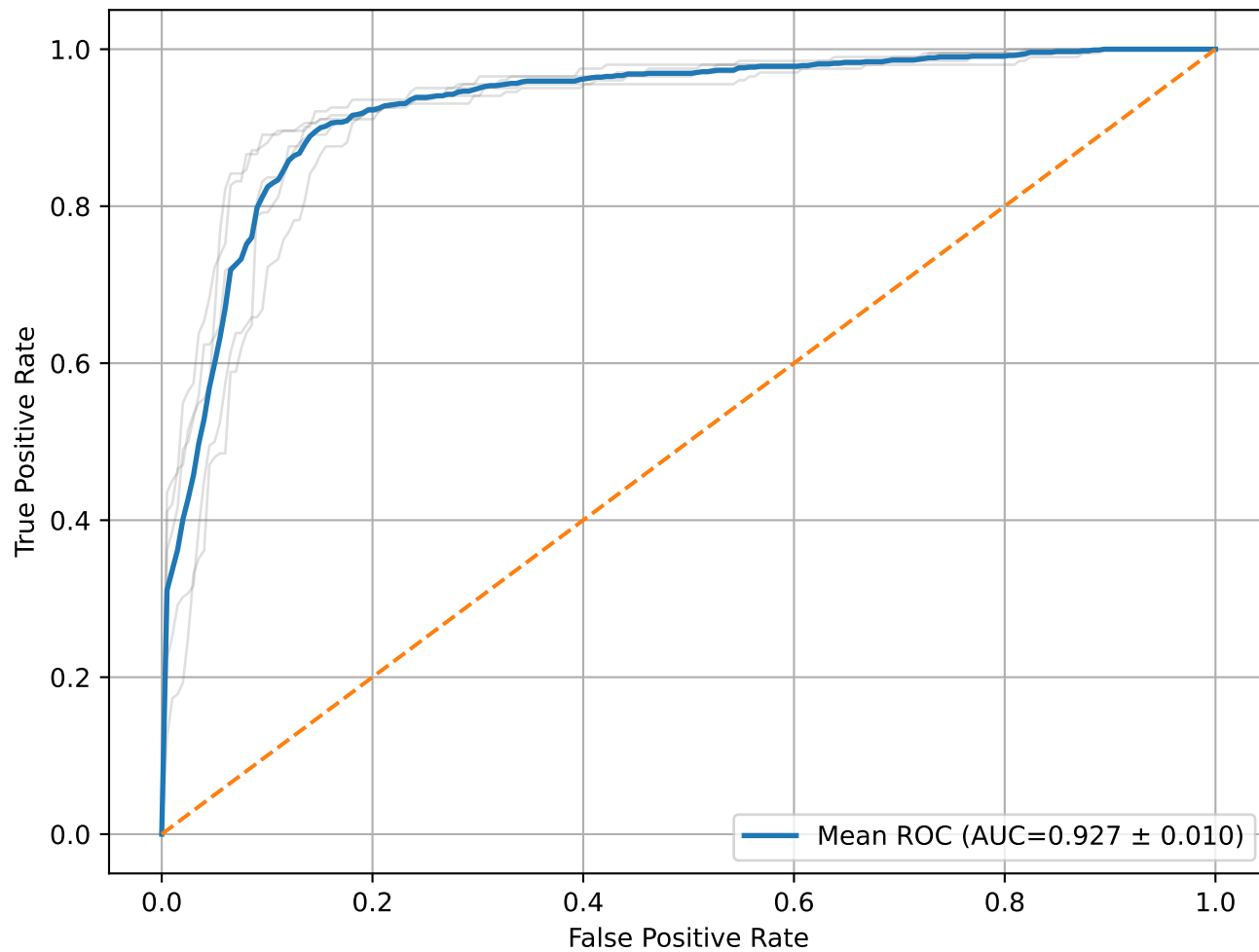
Confusion Matrix - STACKING (last fold)



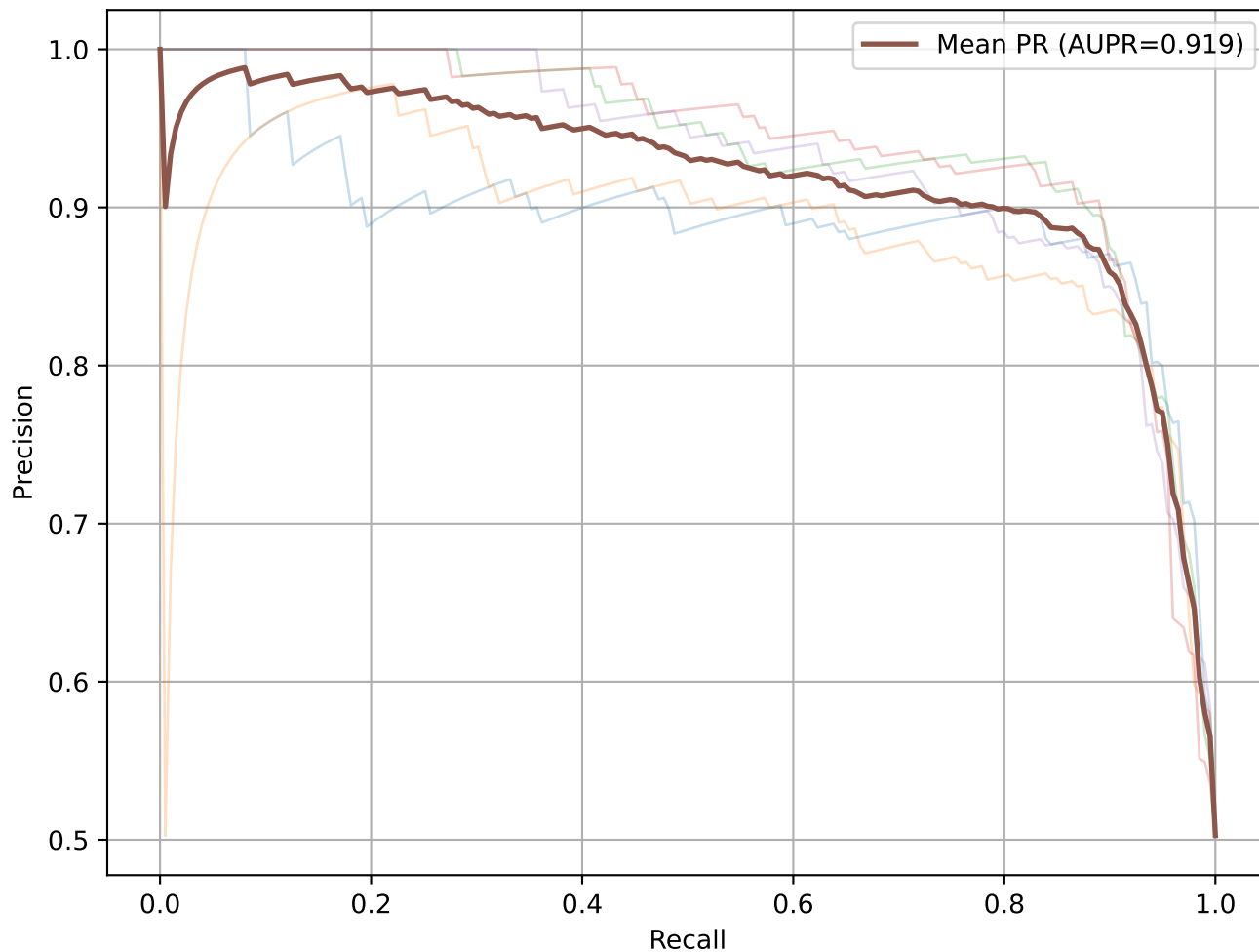
ClassificationReport - STACKING (last fold)

	precision	recall	f1-score	support
0	0.90	0.81	0.85	199
1	0.83	0.91	0.87	202
accuracy			0.86	401
macro avg	0.87	0.86	0.86	401
weighted avg	0.87	0.86	0.86	401

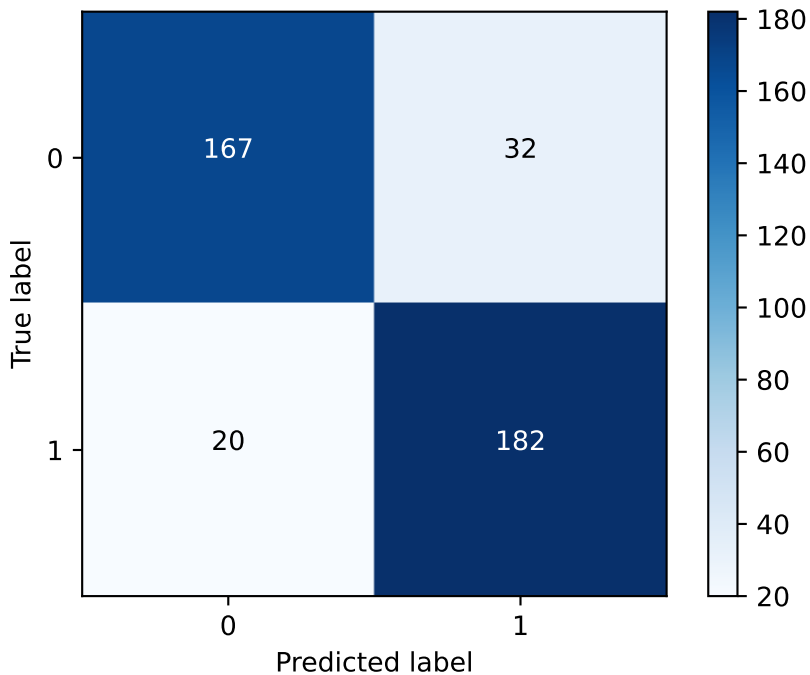
ROC - VOTING



Precision-Recall - VOTING



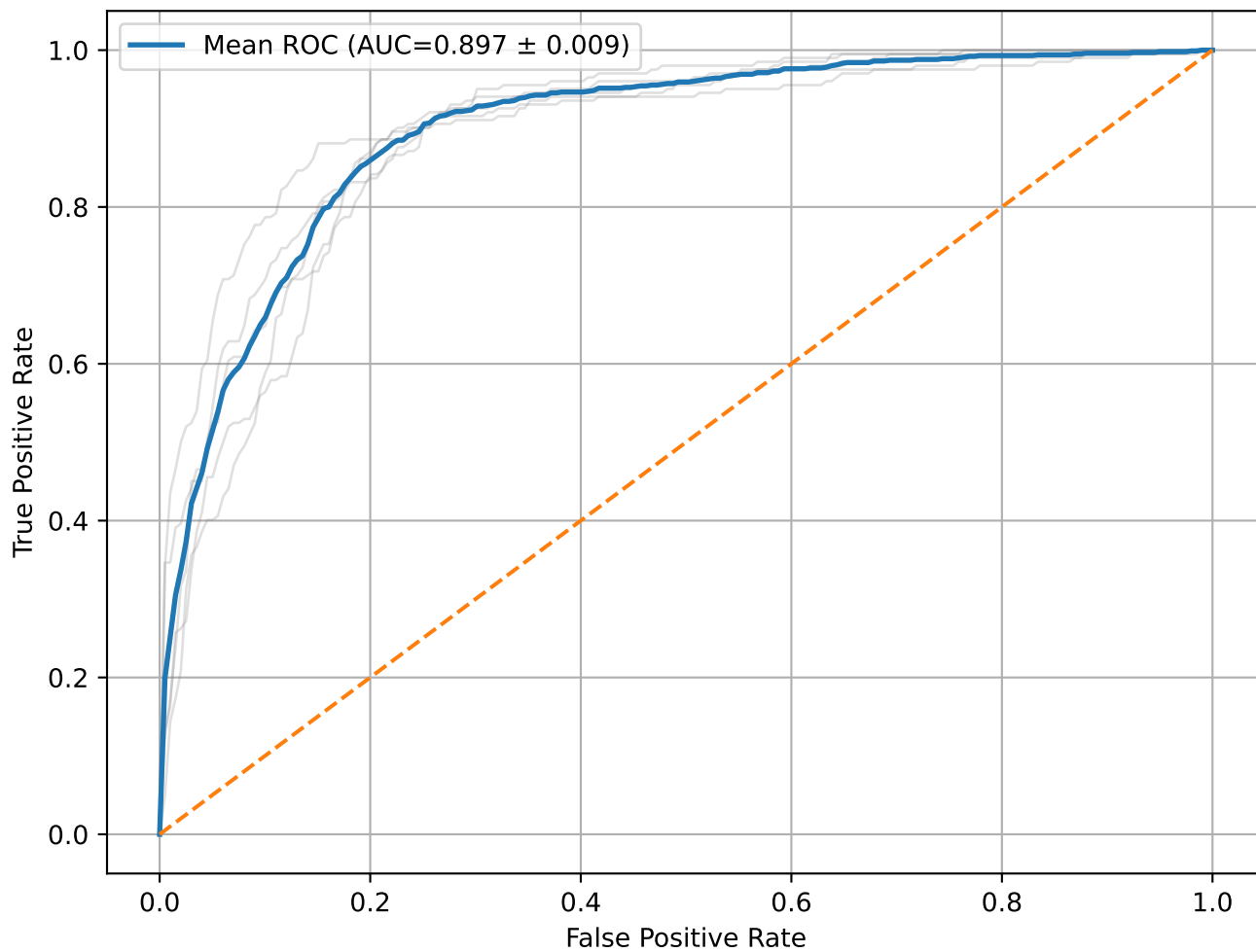
Confusion Matrix - VOTING (last fold)



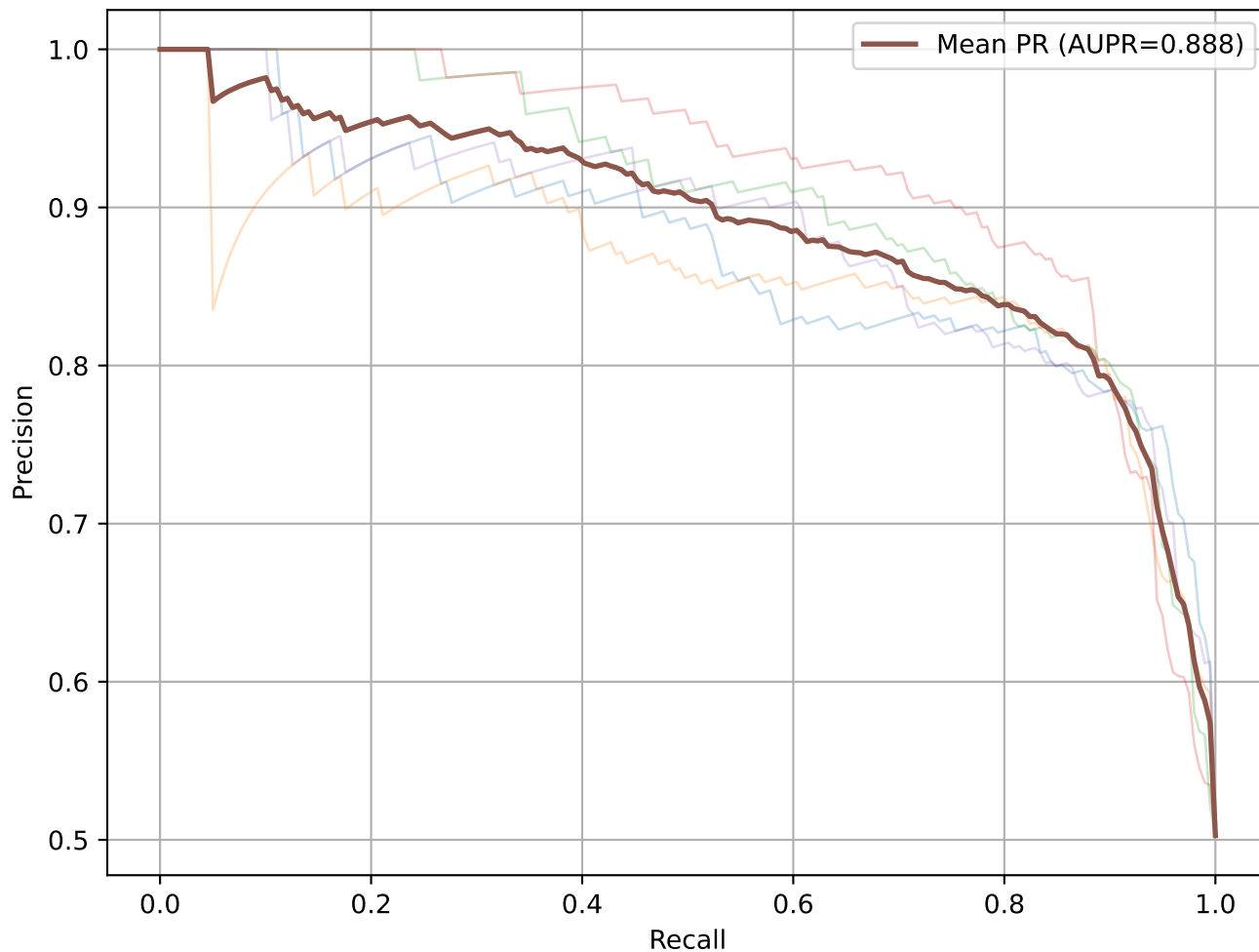
ClassificationReport - VOTING (last fold)

	precision	recall	f1-score	support
0	0.89	0.84	0.87	199
1	0.85	0.90	0.88	202
accuracy			0.87	401
macro avg	0.87	0.87	0.87	401
weighted avg	0.87	0.87	0.87	401

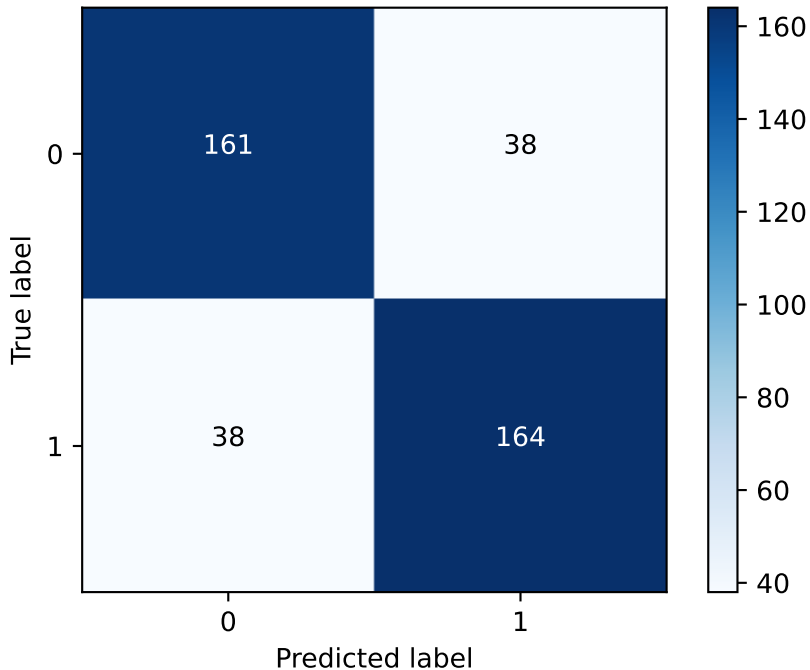
ROC - BAGGING



Precision-Recall - BAGGING



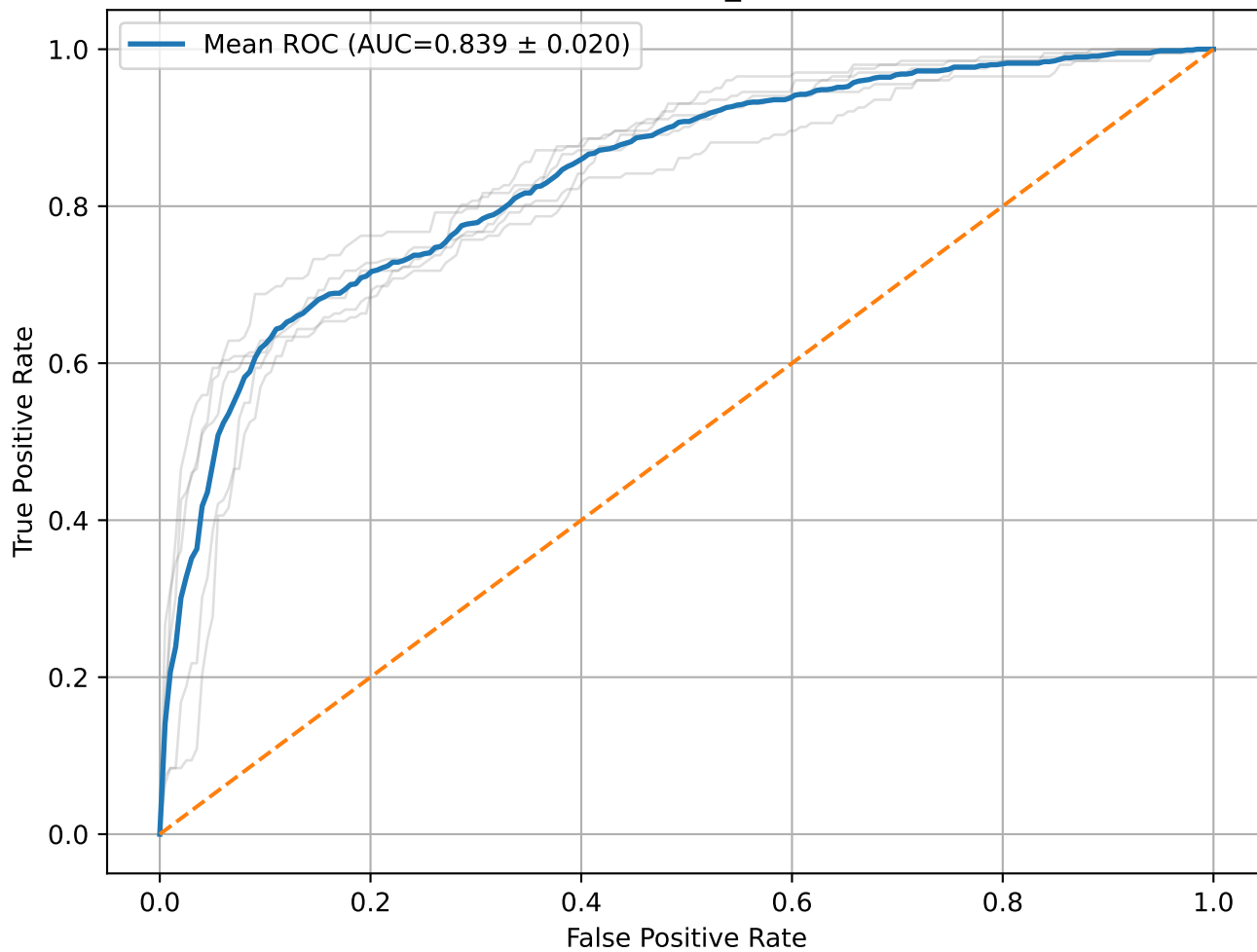
Confusion Matrix - BAGGING (last fold)



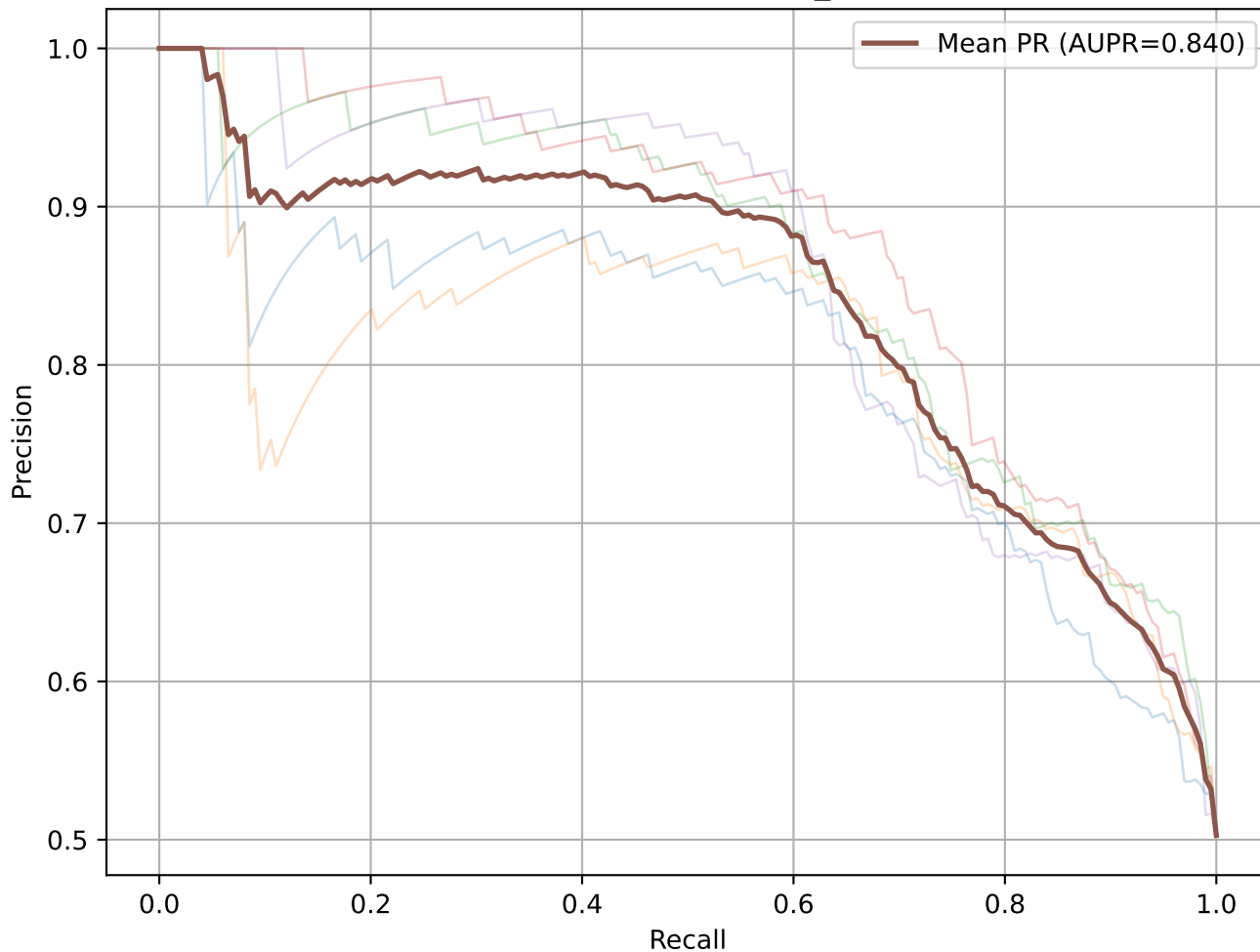
ClassificationReport - BAGGING (last fold)

	precision	recall	f1-score	support
0	0.81	0.81	0.81	199
1	0.81	0.81	0.81	202
accuracy			0.81	401
macro avg	0.81	0.81	0.81	401
weighted avg	0.81	0.81	0.81	401

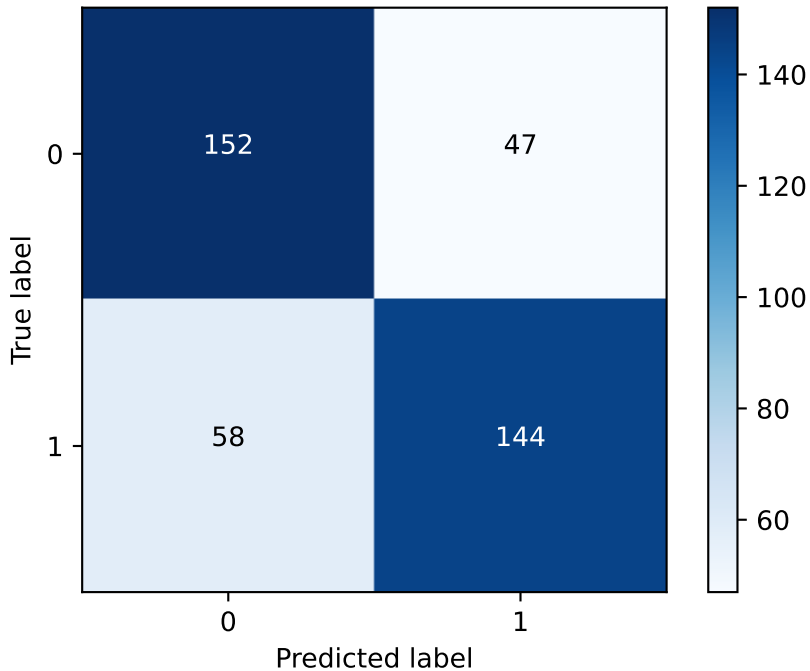
ROC - PCA_SVM



Precision-Recall - PCA_SVM



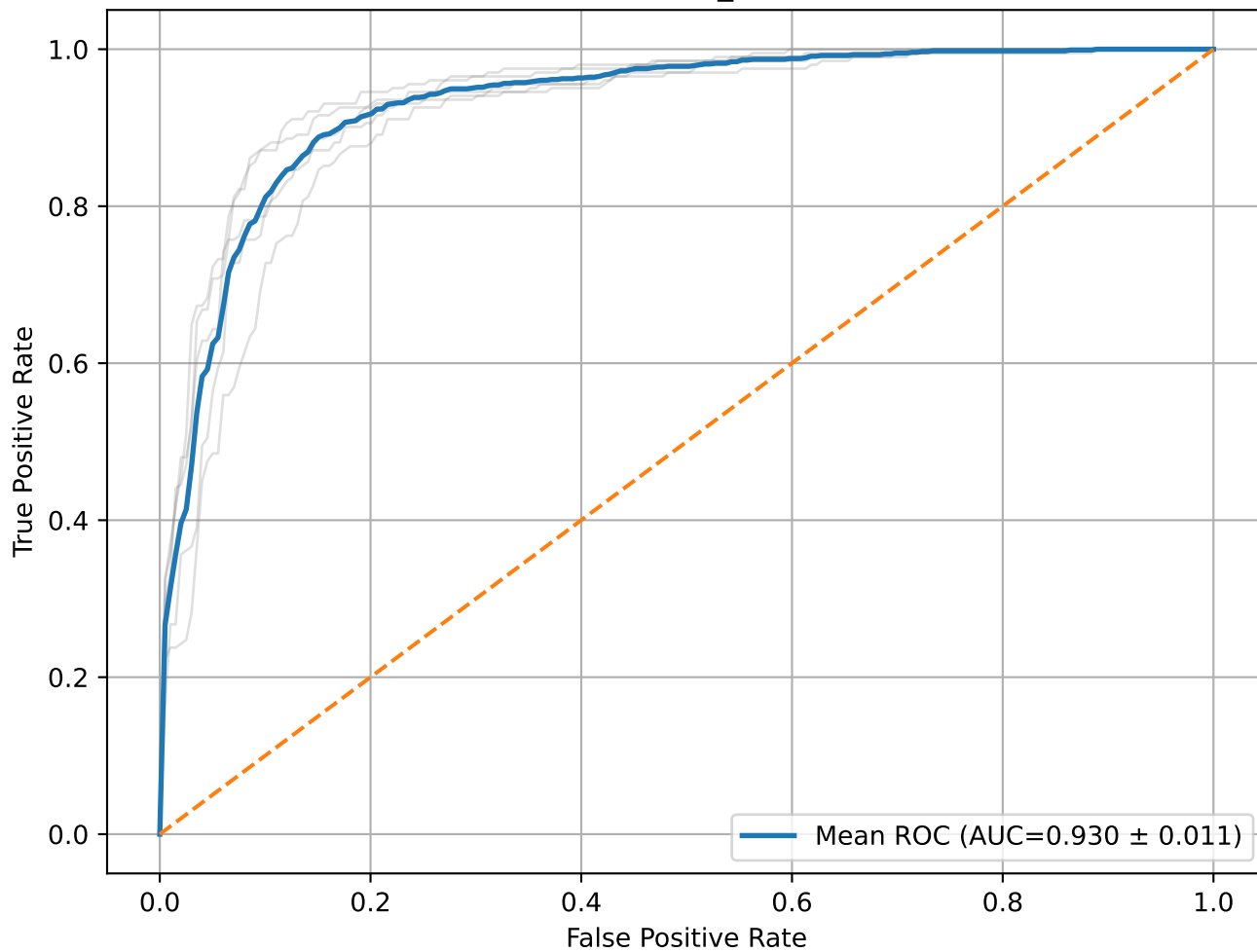
Confusion Matrix - PCA_SVM (last fold)



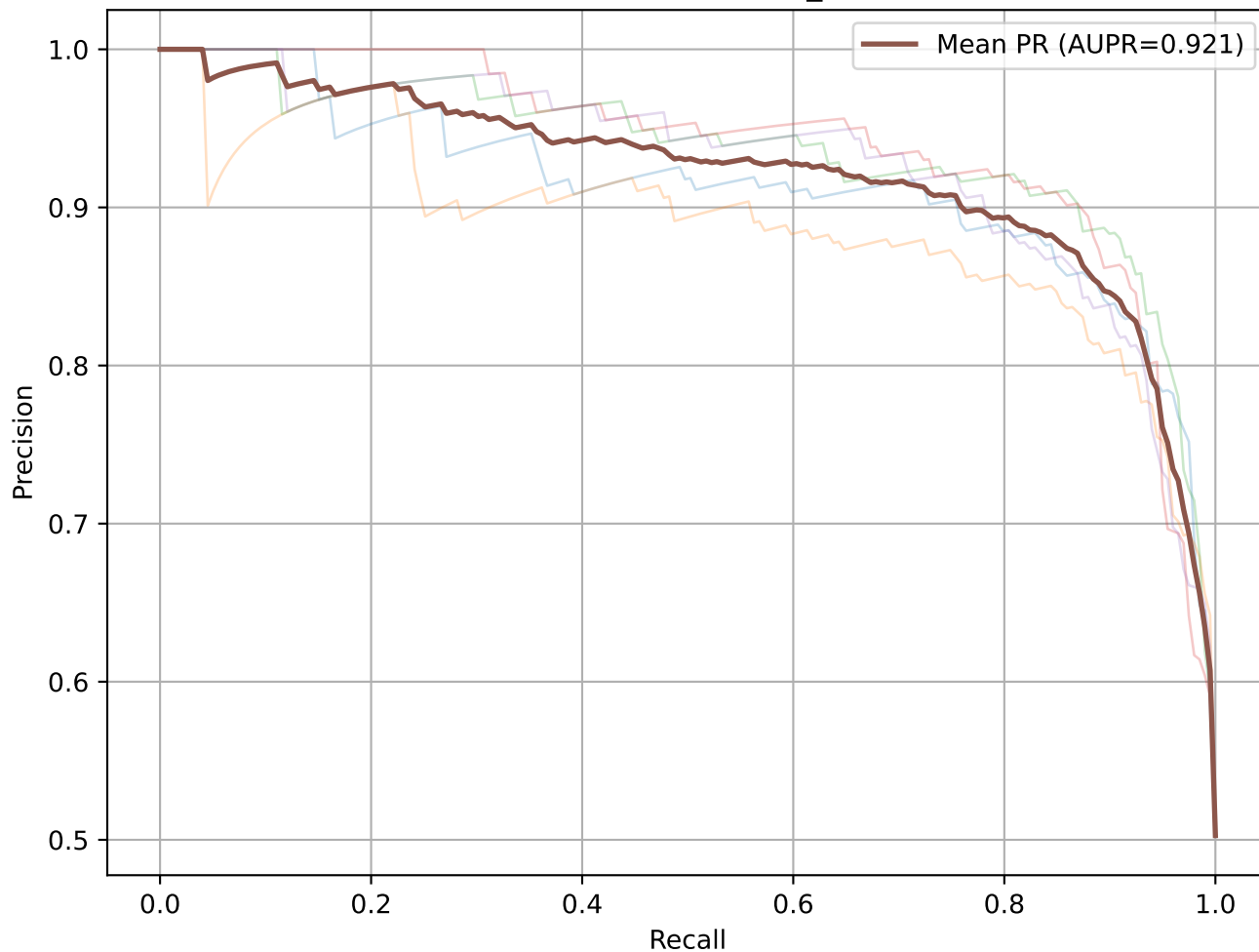
ClassificationReport - PCA_SVM (last fold)

	precision	recall	f1-score	support
0	0.72	0.76	0.74	199
1	0.75	0.71	0.73	202
accuracy			0.74	401
macro avg	0.74	0.74	0.74	401
weighted avg	0.74	0.74	0.74	401

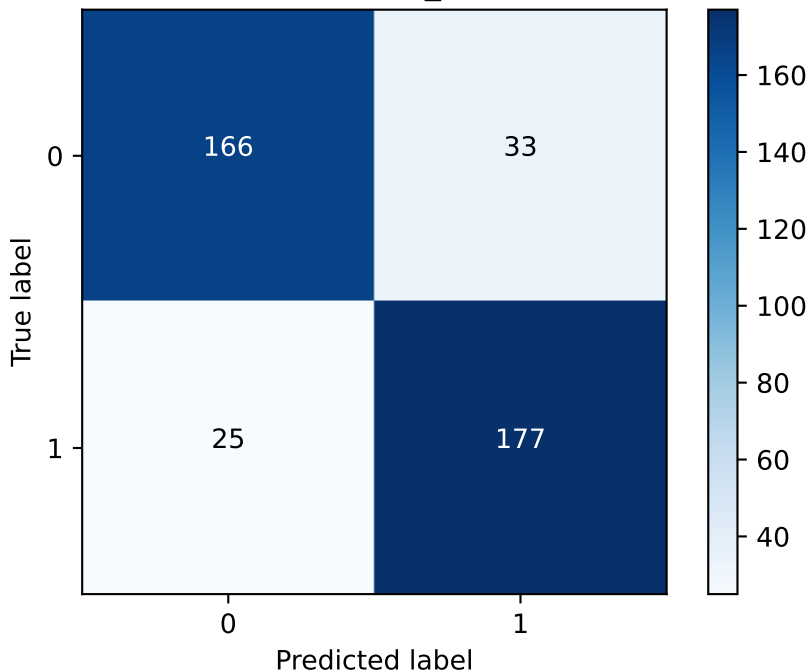
ROC - GA_XGB



Precision-Recall - GA_XGB



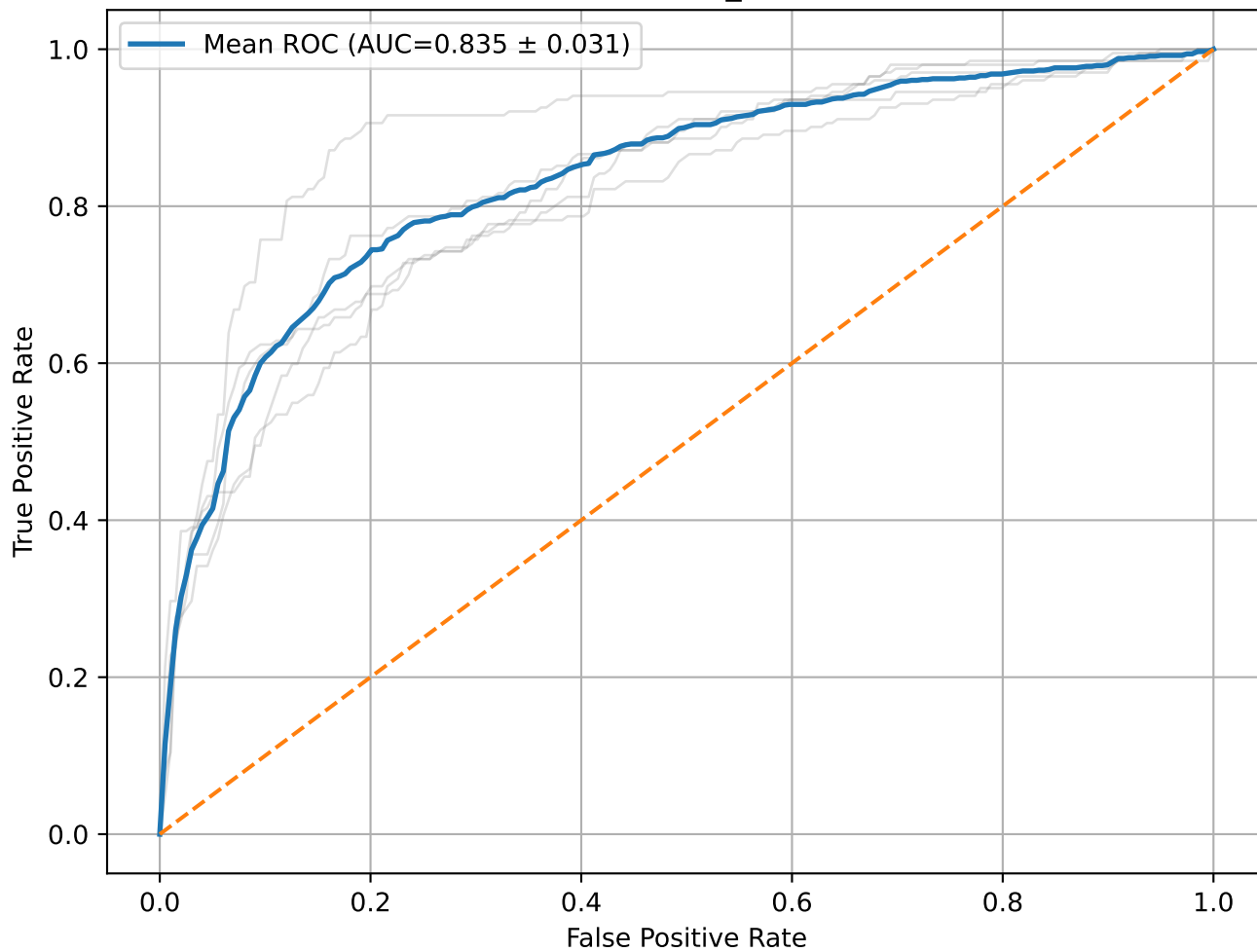
Confusion Matrix - GA_XGB (last fold)



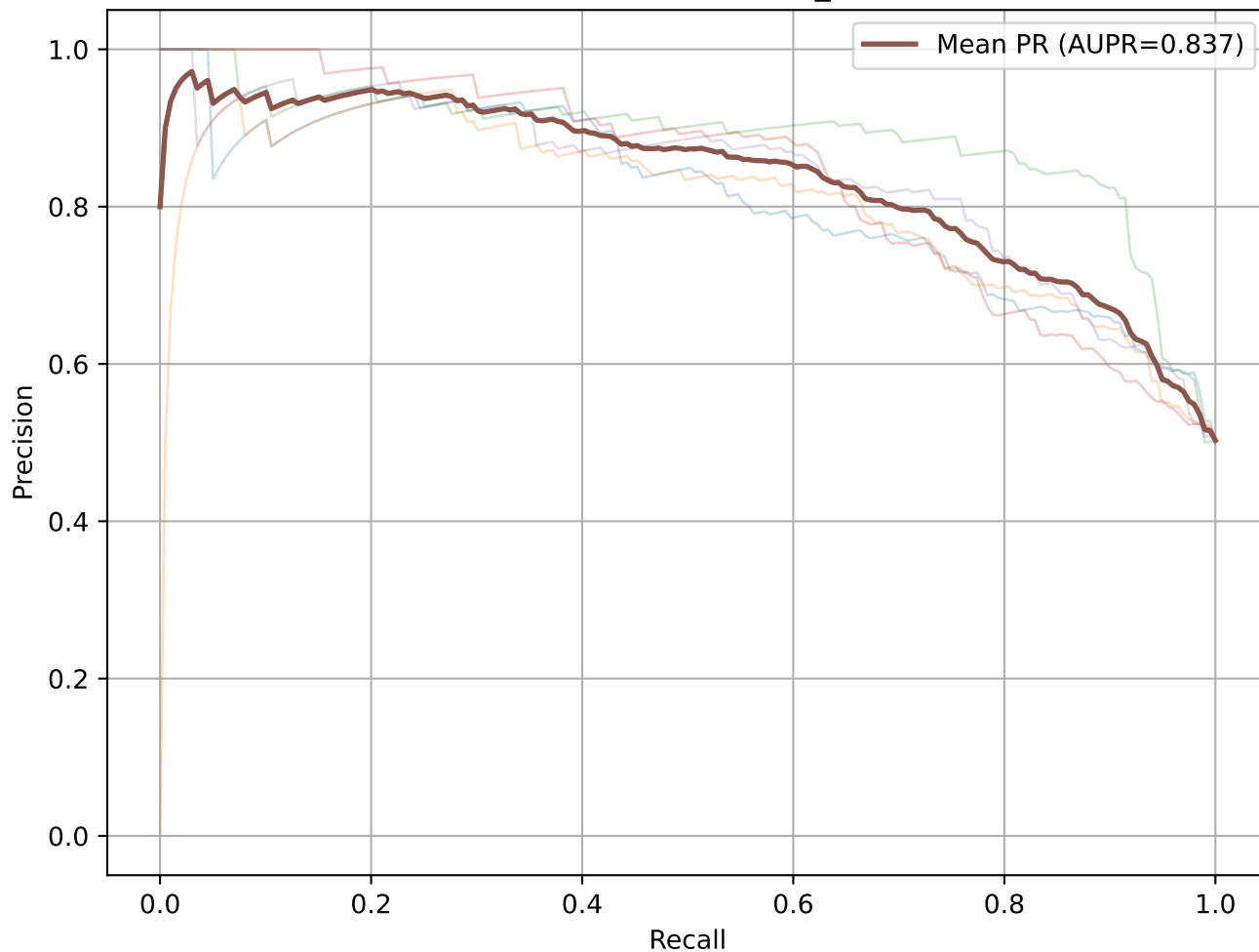
ClassificationReport - GA_XGB (last fold)

	precision	recall	f1-score	support
0	0.87	0.83	0.85	199
1	0.84	0.88	0.86	202
accuracy			0.86	401
macro avg	0.86	0.86	0.86	401
weighted avg	0.86	0.86	0.86	401

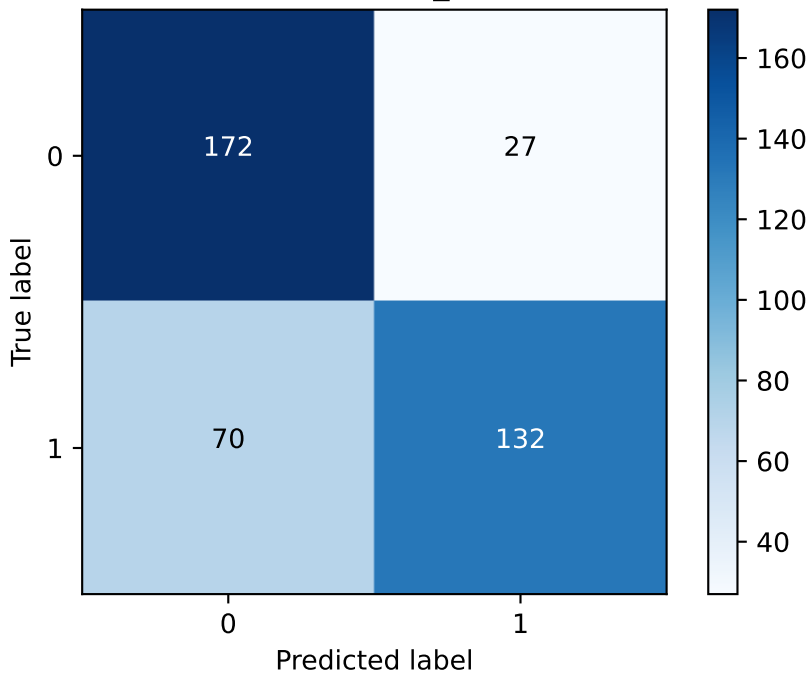
ROC - PSO_SVM



Precision-Recall - PSO_SVM



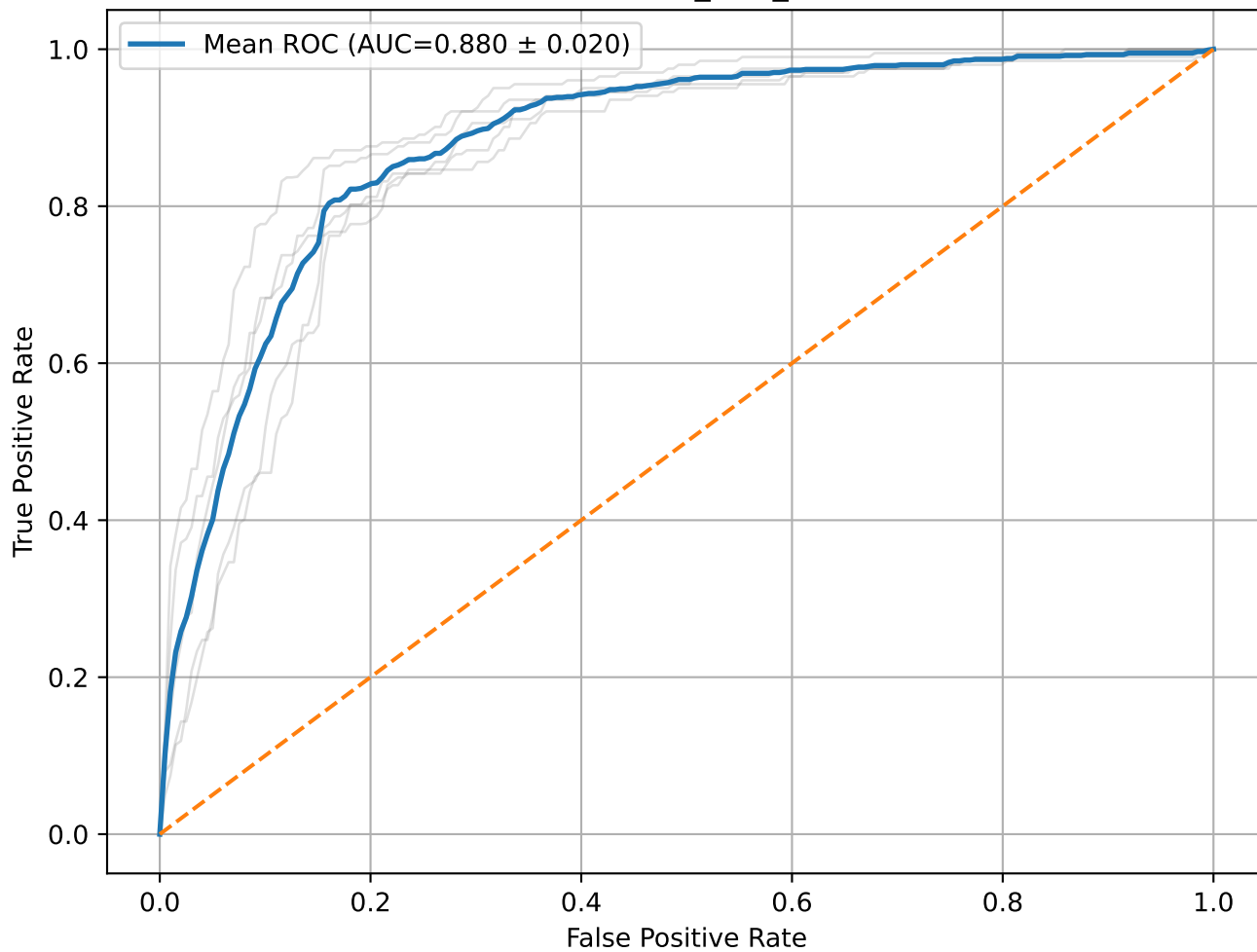
Confusion Matrix - PSO_SVM (last fold)



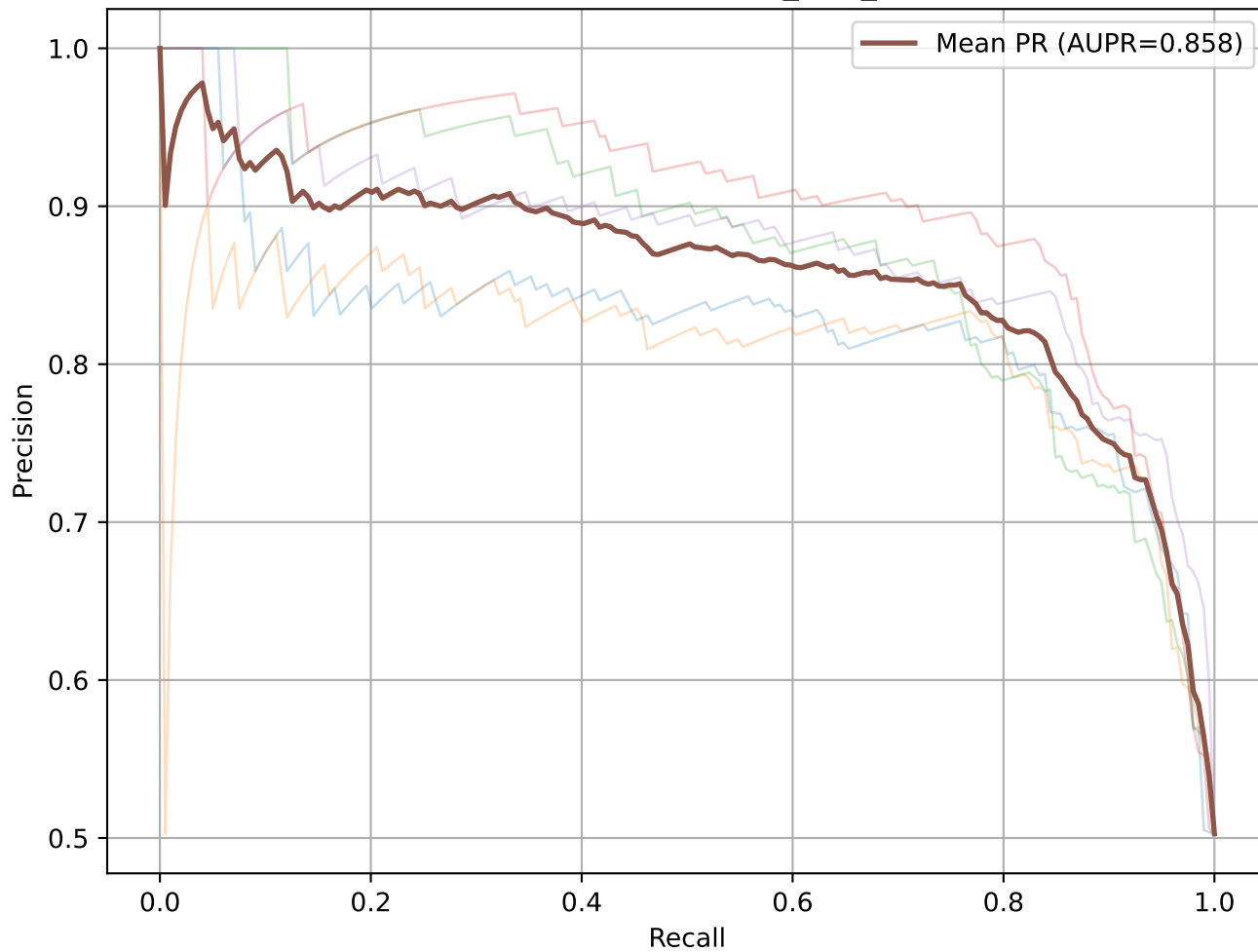
ClassificationReport - PSO_SVM (last fold)

	precision	recall	f1-score	support
0	0.71	0.86	0.78	199
1	0.83	0.65	0.73	202
accuracy			0.76	401
macro avg	0.77	0.76	0.76	401
weighted avg	0.77	0.76	0.76	401

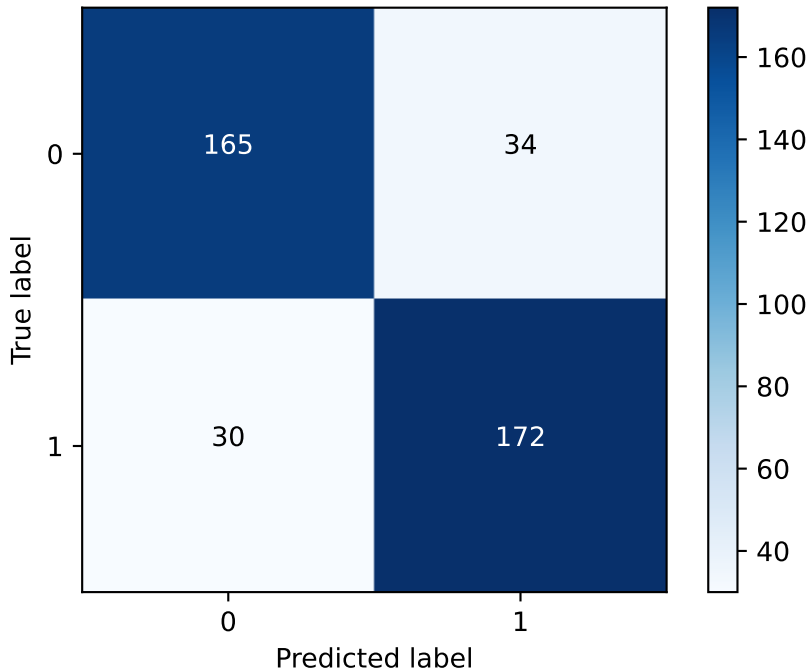
ROC - BAYES_OPT_SVMS



Precision-Recall - BAYES_OPT_SVMS



Confusion Matrix - BAYES_OPT_SVMS (last fold)



ClassificationReport - BAYES_OPT_SVMS (last fold)

	precision	recall	f1-score	support
0	0.85	0.83	0.84	199
1	0.83	0.85	0.84	202
accuracy			0.84	401
macro avg	0.84	0.84	0.84	401
weighted avg	0.84	0.84	0.84	401

Model Comparison Table (sorted by mean ROC AUC)

model	mean_roc_auc	std_roc_auc	mean_aupr	mean_accuracy	mean_f1
lgb	0.932865	0.009626	0.923558	0.867101	0.869470
xgb	0.928182	0.008079	0.919865	0.870081	0.873561
ga_xgb	0.928083	0.011027	0.920151	0.865600	0.868744
voting	0.925393	0.009847	0.918461	0.875557	0.879039
rf	0.924405	0.010962	0.921893	0.875562	0.877619
stacking	0.919419	0.011097	0.908224	0.868091	0.874074
bagging	0.895835	0.009312	0.888124	0.826771	0.826463
logistic	0.886831	0.023555	0.867440	0.818819	0.820857
bayes_opt_svms	0.878013	0.020496	0.857818	0.817334	0.820526
dnn	0.871025	0.013257	0.875649	0.803883	0.796121
svm	0.853846	0.036245	0.849592	0.788437	0.778959
pca_svm	0.837811	0.020364	0.840097	0.755591	0.744878
pso_svm	0.833793	0.030794	0.836634	0.764556	0.743043

Combined ROC - All Models

